

A
Summer Internship Report
on
(UPCOMING PROJECTS)
Completed
at
POWER GRID CORPORATION OF INDIA LIMITED, SRIKONA



Project Name

Preparation of Quantity Estimates, Bar Bending Schedule (BBS), and Bill of Quantities (BOQ) for Civil Works to Facilitate Technical and Administrative Approval at the 132/400 kV PGCIL Srikona Substation.

Duration

25 May to 8 July, 2026

Submitted

By

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DEPARTMENT OF CIVIL ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY, SILCHAR

ACKNOWLEDGMENT

We, the students of **civil engineering** at the **National Institute of Technology Silchar**, are extremely grateful for the opportunity to undertake our industrial internship at **Power Grid Corporation of India Limited**.

I am deeply grateful to **Mr. Aninda Deb Laskar sir (DGM, POWERGRID Silchar SS)** and **Mr. Raju Sarkar (JE, POWERGRID Silchar SS)** for their valuable guidance, constant supervision, and technical insights throughout the internship period. Their support has been instrumental in bridging the gap between theoretical knowledge and field practices in civil engineering.

We further extend our gratitude to the **Head of the Department** and **The Training and Placement Cell** of the **National Institute of Technology Silchar** for their continuous support and for facilitating this internship opportunity.

This training has provided us with hands-on exposure to the practical aspects of civil engineering, complementing our academic learning and helping us grow as future professionals in the field.

SL NO.	NAME	REGISTRATION NO.	SIGNATURE
1.	Vaishali Dhar	2311098	
2.	Aparna Singh	2311020	

DECLARATION

I hereby declare that the "**45 days internship report**" is a genuine and original record of the work carried out by me during the period from **May 25, 2026 to July 8, 2026** as part of the summer internship training required for the fulfillment of the Bachelor of Technology (B.Tech) degree in civil engineering at the National Institute of Technology Silchar.

The training was undertaken under the supervision and guidance of the **Mr. Raju Sarkar, Junior Engineer at Power Grid Corporation of India Limited.**

I also confirm that this report is an original submission and has not been presented elsewhere for the fulfillment of any academic qualification or certification.

SL NO.	NAME	REGISTRATION NO.	SIGNATURE
1.	Vaishali Dhar	2311098	
2.	Aparna Singh	2311020	

.....
MR. RAJU SARKAR
(JUNIOR ENGINEER AT PGCIL)

.....
MR. ANINDA DEB LASKAR
(DGM AT PGCIL)

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1. INTRODUCTION

Power Grid Corporation of India Limited (POWERGRID), a Government of India Enterprise under the Ministry of Power, is the Central Transmission Utility (CTU) of the country, responsible for planning, implementing, operating and maintaining the Inter-State Transmission System (ISTS). Civil engineering works have been undertaken for four upcoming infrastructure projects. These include the construction of an underground bunker as part of the disaster preparedness plan to safeguard essential personnel and critical equipment during emergencies such as natural disasters or hostile situations. Accessibility improvements have been incorporated by providing ramps at the Control Room and Transit Camp entrances to facilitate the movement of wheelchair users and persons with disabilities. Bamboo crib structures have been proposed around vulnerable transmission tower locations to mitigate soil erosion and enhance slope stability. In addition, the structural design of a crane shed has been prepared to facilitate the safe arrival, parking, and maintenance of cranes required for operational and maintenance activities.

This report presents the work carried out during a summer internship at POWERGRID, covering the design interpretation, Quantity Estimation, Bar Bending Schedule (BBS) and preparation of Bill of Materials (BOM) estimation of all the above civil works. The project provided practical exposure to reinforced cement concrete (RCC) detailing, quantity surveying, and the tender-stage documentation process followed for critical government infrastructure projects.

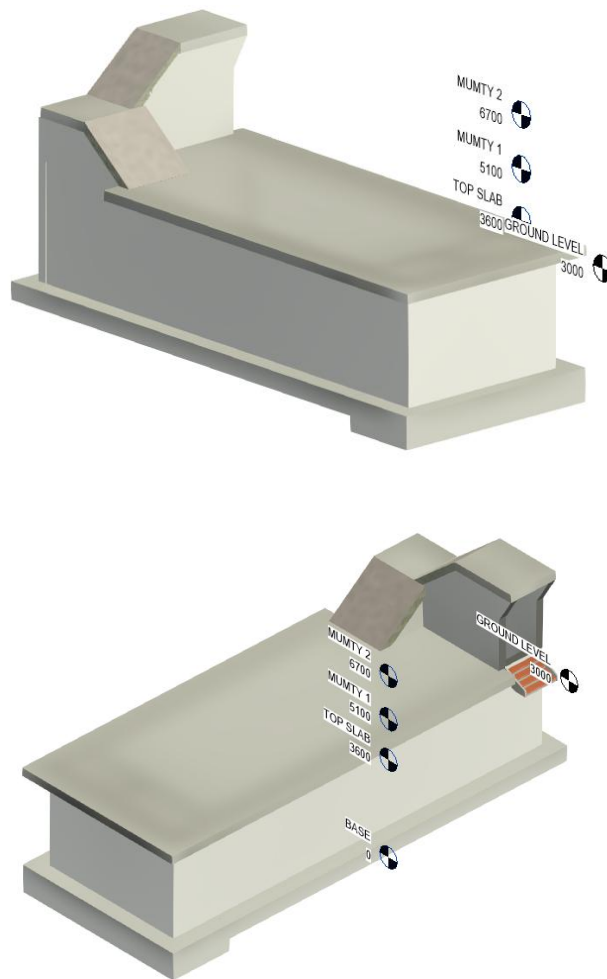
The construction of the bunker was carried out in accordance with the relevant Bureau of Indian Standards (BIS) codes, including:

- **IS 456:2000** — Plain and Reinforced Concrete — Code of Practice

- **IS 875 (Parts 1–5)** — Design Loads (Other than Earthquake) for Buildings and Structures
- **IS 1893:2016** — Criteria for Earthquake Resistant Design of Structures
- **IS 2502** — Code of Practice for Bending and Fixing of Bars for Concrete Reinforcement
- **IS 13920** — Ductile Detailing of RCC Structures Subjected to Seismic Forces
- **CPWD Delhi Schedule of Rates (DSR) 2023** — for cost estimation and BOQ preparation
- **CPWD Specifications 2019** — India’s standard guidelines for quality, execution and billing in government construction.

Among these, **IS 10262** is particularly significant, offering detailed guidance for concrete mix design using local materials. It helps ensure optimal strength, workability, durability, and cost-efficiency-vital for major infrastructure projects.

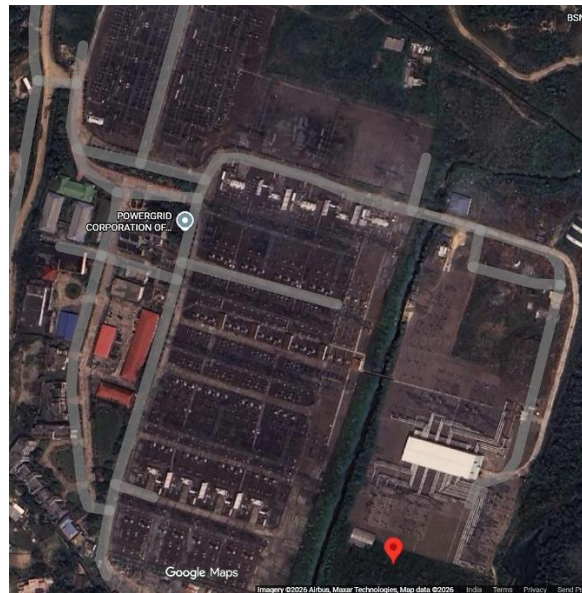
2. ABOUT THE PROJECT



Construction of a reinforced cement concrete (RCC) protective bunker at PGCIL, Srikona, for safeguarding personnel during emergency situations arising from security contingencies in the adjoining international border area, including all civil, structural, waterproofing, finishing, and allied works complete as per approved drawings, technical specifications, and the directions of the Engineer-in-Charge.

The bunker is a partially below-ground RCC structure comprising a bottom (raft) slab founded at (-)3.15 m level, a mid-level top slab at (+)0.6 m, and a mumty (access superstructure) rising up to (+)3.15 m level, connected to the ground surface through a dedicated staircase, with an attached toilet/bath block and septic tank/soak-pit drainage arrangement.

2.1 SITE LOCATION AND CONNECTIVITY



The geographic and logistical placement of the site plays a critical role in its construction supply chain (procurement of aggregate, cement, and steel).

- **Latitude & Longitude:** 24°49'34.0"N 92°43'19.8"E
- **Location:** 400/132 kV POWERGRID Silchar Substation, Srikona, Cachar, Assam
- **Connectivity:** The site is connected directly to two lane National Highway (NH 37) that serves as a strategic transportation corridor linking Imphal (Manipur) to Silchar and Sribhumi in Assam, via Lalmati TV Tower Road at Indragarhgrant, ensuring all weather efficient connectivity. Its proximity to major transit corridors ensures the unhindered movement of heavy machinery, Transit Mixers, and Ready Mix Concrete (RMC) vehicles.

PROJECT NAME:	Construction of Underground Bunker in Critical Switchyard/Establishment under Disaster Preparedness of Transmission Infrastructure/Employees
CLIENT:	Power Grid Corporation of India Limited (A Government of India Enterprise)

ARCHITECTURAL DRAWING NO:	C/ENGG/TIG/BUNKER/ARCH/01 (Sheets 1–3)
STRUCTURAL DRAWING NO:	C/ENGG/TIG/BUNKER/STR/01 (Sheets 1–6 + Mumty & Detail sheets)
STRUCTURAL TYPE:	Underground RCC framed bunker with raft foundation
OVERALL PLAN DIMENSION (BOTTOM SLAB):	$\approx 16.44 \text{ m} \times 6.44 \text{ m}$
BOTTOM SLAB LEVEL:	(-) 3.15 m LVL TOC (bunker floor); (-) 2.70 m LVL TOC (adjoining footing zone)
TOP SLAB LEVEL:	(+) 0.60 m LVL TOC
MUMTY SLAB LEVEL:	(+) 1.55 m and (+) 3.15 m LVL TOC
ACCESS:	Dog-legged RCC staircase (Flight 1 & Flight 2) with landings
ANCILLARY FACILITIES:	Toilet/Bath block, Septic Tank, Manhole (as per Municipal Corporation Delhi design, Clause 19.4)

2.2 STRUCTURAL DESCRIPTION & SPECIFICATIONS

3.1 STRUCTURAL COMPONENTS

-----BOTTOM (RAFT) SLAB-----

Provided at (-)2.7 m and (-)3.15 m LVL TOC, acting as the foundation raft for the entire bunker footprint (approx. 16.44 m × 6.44 m); reinforced with 10 mm and 12 mm dia bars at 150–175 mm c/c.

-----WALLS-----

RCC walls (350 mm thick) at outer faces reinforced with 16 mm dia vertical bars @ 175 mm c/c on the outer face and 12 mm dia bars @ 175 mm c/c on the inner face, with 10 mm dia horizontal distribution bars.

-----COLUMNS-----

RCC columns of 300 × 300 mm with 8 nos. 12 mm dia longitudinal bars and 8 mm dia ties/stirrups @ 190 mm c/c (Section X-X).

-----BEAMS-----

Tie beams TB1 (300 × 450 mm) and TB2 (300 × 600 mm) with 20–25 mm dia main bars and 10 mm dia stirrups @ 150 mm c/c; Main Beam (MB) and Secondary Beam (SB) of 300 × 450 mm at mumty roof level.

-----TOP SLAB-----

Flat slab at (+)0.6 m LVL TOC near the entry gate and at the rear of the entry gate, reinforced with 12 mm dia bars @ 150 mm c/c (top & bottom).

-----MUMTY SLAB-----

Provided at (+)1.55 m and (+)3.15 m LVL TOC with 12 mm dia bars @ 150 mm c/c (top & bottom), forming the roof over the staircase head and toilet block.

-----STAIRCASE-----

Dog-legged RCC staircase (Landing 1 + Flight 1 + Landing 2, and Flight 2) reinforced with 12 mm dia main and distribution bars @ 150 mm c/c.

-----TOP SLAB-----

Toilet/Bath modules of size 1.21 m × 2.12 m each, connected to a septic tank and municipal-type manhole (as per MCD design, Clause 19.4) for effluent disposal.

3.2 FINISHING AND SERVICES

-----FLOORING-----

Cement concrete (1:2:4) flooring in the bunker and staircase; anti-skid vitrified tiles (600 × 600 mm) in the toilet/bath area.

-----WALL FINISHING-----

Wall finishing: ceramic glazed wall tiles as dado up to 2100 mm in the toilet area; premium acrylic smooth exterior paint; acrylic emulsion paint internally.

-----DOORS AND VENTILATION-----

Doors and ventilation: uPVC glazed doors for the toilet, an M.S. sheet entry door with angle-iron frame, and louvered aluminium/M.S. ventilators fitted with fly-proof SS jali.

-----PLUMBING AND SANITARY -----

Plumbing and sanitary: CPVC water supply piping, uPVC SWR drainage piping, and sanitary fixtures such as WC, wash basin, health faucet, geyser, and CP fittings.

-----ELECTRICAL -----

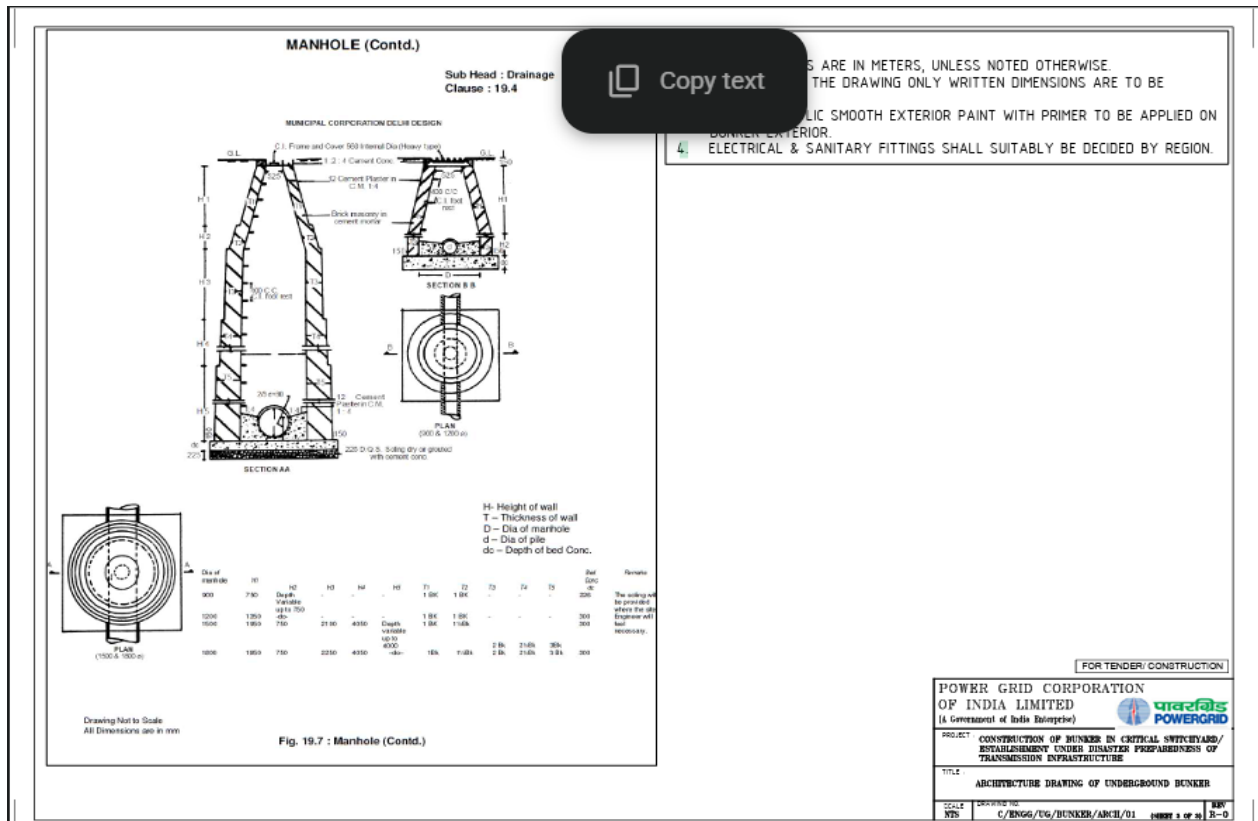
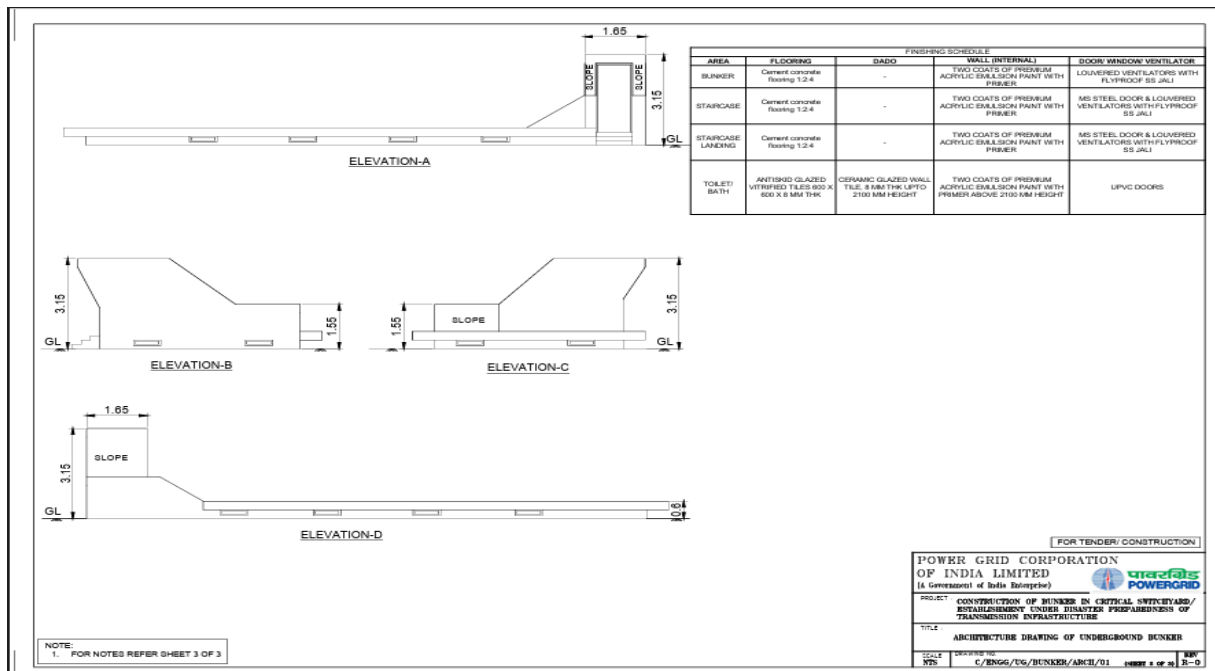
Internal wiring for light/fan points, MCB distribution, LED light fittings, exhaust fans, and wall fans.

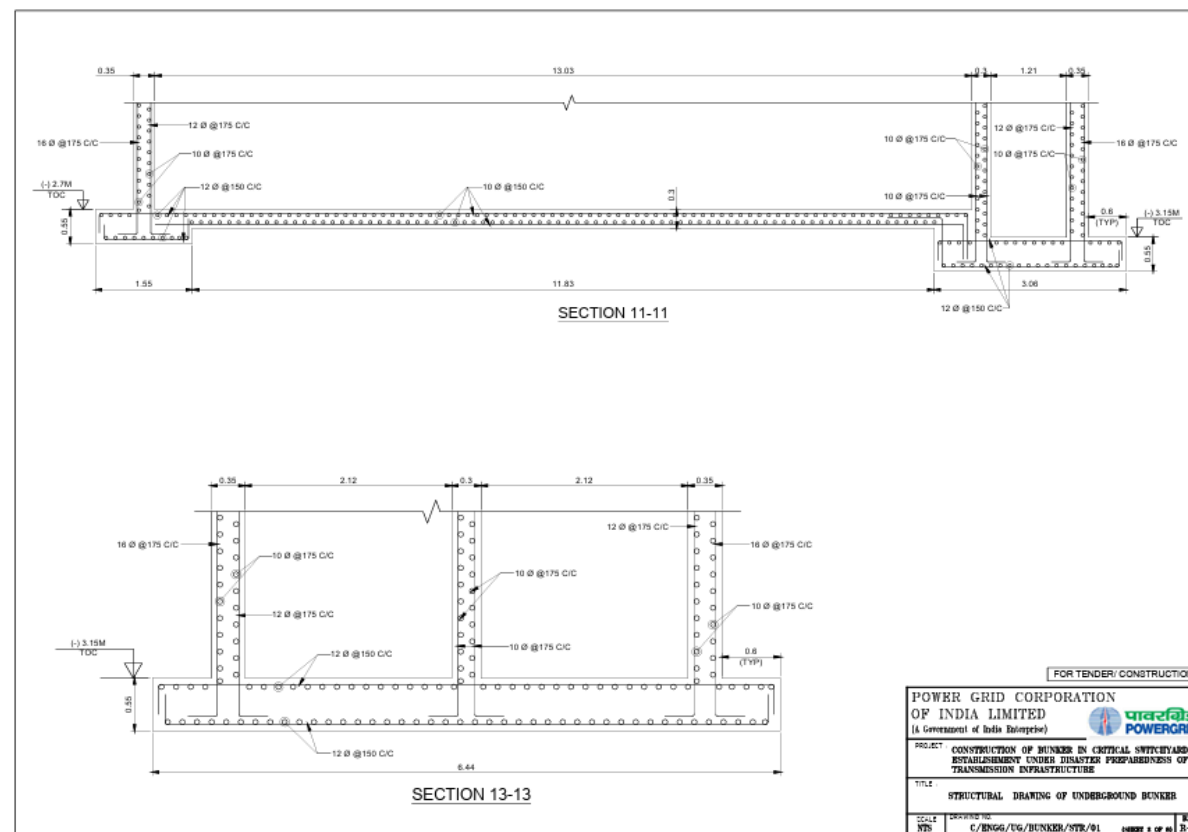
AREA	FLOORING	DADO	WALL (INTERNAL)	DOOR/WINDOW/VENTILATION
BUNKER AREA	Cement concrete flooring 1:2:4	-	Two coats premium acrylic emulsion paint with primer	MS steel door & louvered ventilators with flyproof SS Jali

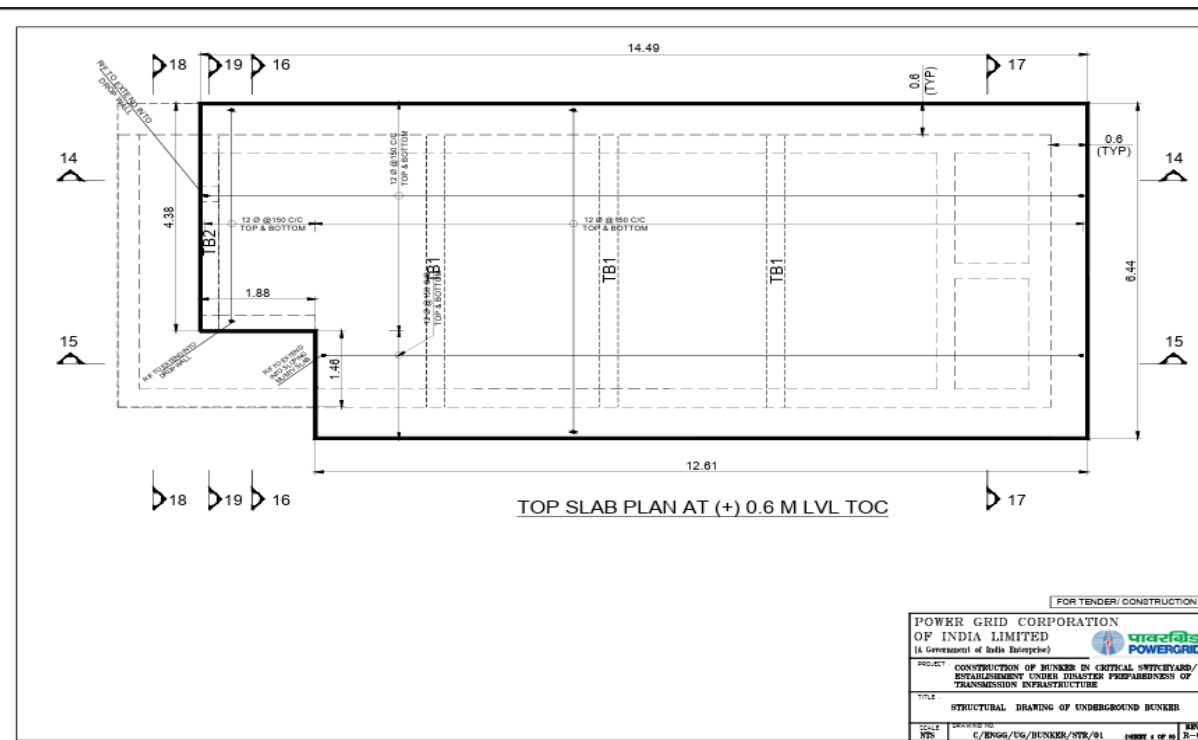
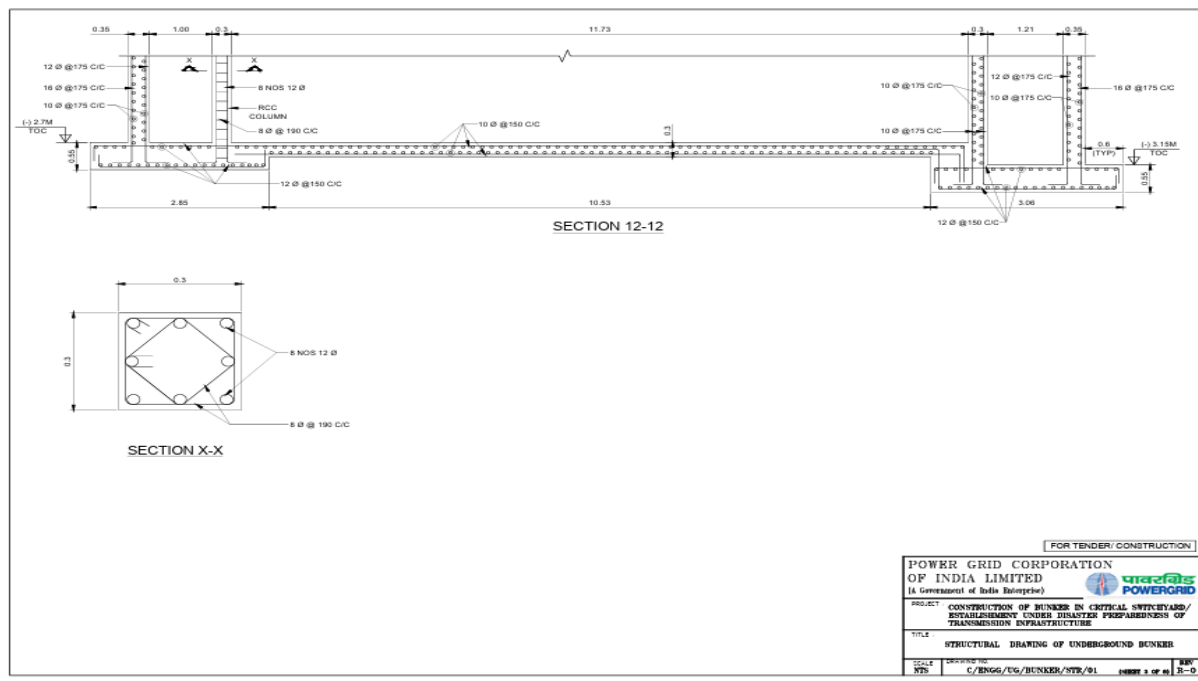
AREA	FLOORING	DADO	WALL (INTERNAL)	DOOR/WINDOW/VENTILATION
STAIRCASE	Cement concrete flooring 1:2:4	-	Two coats premium acrylic emulsion paint with primer	MS steel door & louvered ventilators with flyproof SS Jali
STAIRCASE LANDING	Cement concrete flooring 1:2:4	-	Two coats premium acrylic emulsion paint with primer	Louvered ventilators with flyproof SS Jali
TOILET/BATH	Antiskid glazed vitrified tiles, 600 × 8 mm thk	Ceramic glazed wall tile, 8 mm thk, up to 2100 mm height	Two coats premium acrylic emulsion paint with primer	uPVC doors

2.3 ARCHITECTURAL & STRUCTURAL DRAWINGS

The following pages reproduce the key architectural and structural drawings referred to during the internship for the preparation of its **Quantity Estimation, the Bar Bending Schedule (BBS) and Bill of Quantities (BOQ)**.





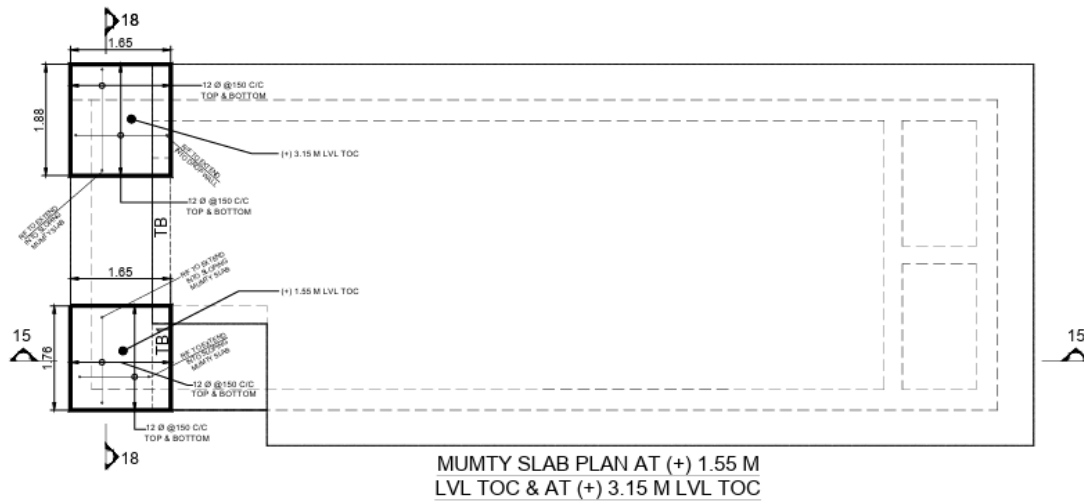


This diagram shows a cross-section of a wall and floor assembly at a door threshold. The wall has a total thickness of 12 inches, composed of two layers of 6-inch concrete masonry units (CMU) separated by a vertical cavity. A horizontal reinforcement bar, labeled "12 @ 150 C/C", runs through the wall. The floor consists of a 4-inch thick concrete slab supported by a steel beam. The beam is reinforced with "12 @ 150 C/C". The floor slab is shown with a top layer of reinforcement. The door threshold is indicated by a dashed line. Dimensions include a 0.37 inch gap between the wall and the floor slab, a 1.1 inch gap between the floor slab and the ceiling, and a 0.02 inch gap between the wall and the ceiling. The ceiling is labeled "LVL VARIES". The floor slab is labeled "(4) 0.6 M T/C". The wall is labeled "12 @ 150 C/C". The floor slab is labeled "4" and "2.12".

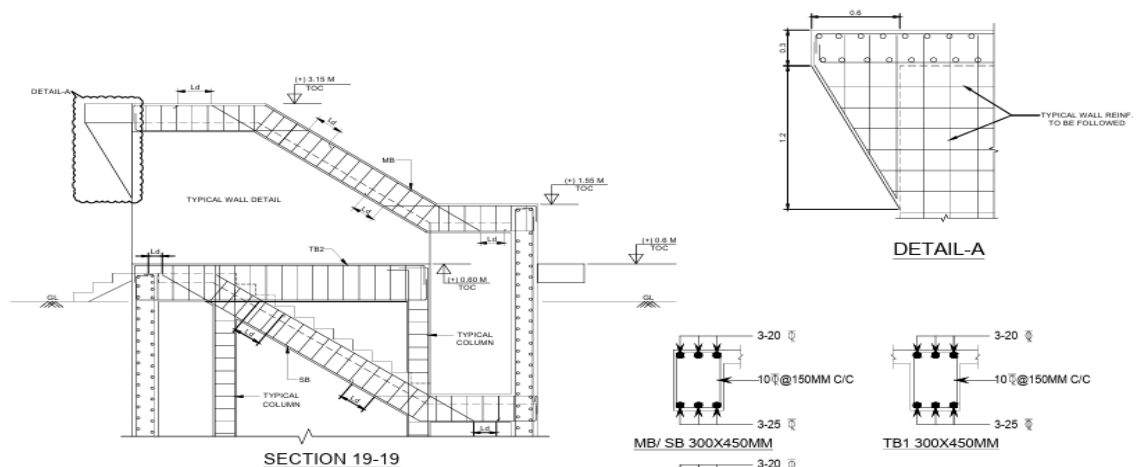
POWER GRID CORPORATION OF INDIA LIMITED (A Government of India Enterprise)		 पावरग्रिड POWERGRID	
PROJECT :	CONSTRUCTION OF BUNKER IN CRITICAL SWITCHYARD/ ESTABLISHMENT UNDER DISASTER PREPAREDNESS OF TRANSMISSION INFRASTRUCTURE		
TITLE :	STRUCTURAL DRAWING OF UNDERGROUND BUNKER		
SCALE :	DRAWING TITLE :		REV :
NTP	C/RINGG/UG/BUNKER/STR/01		R-0



POWER GRID CORPORATION OF INDIA LIMITED (A Government of India Enterprise)		 पावरग्रिड POWERGRID
PROJECT	CONSTRUCTION OF BUNKER IN CRITICAL SWITCHYARD/ ESTABLISHMENT UNDER DISASTER PREPARATIONNESS OF TRANSMISSION INFRASTRUCTURE	
TITLE	STRUCTURAL DRAWING OF UNDERGROUND BUNKER	
SCALE	DRAWING TITLE C/ENGG/UG/BUNKER/STR/01	SHEET NO. OF 04 REV R-0



FOR TENDER/ CONSTRUCTION			
POWER GRID CORPORATION OF INDIA LIMITED (A Government of India Enterprise)			
PROJECT	CONSTRUCTION OF BUNKER IN CRITICAL SWITCHYARD/ ESTABLISHMENT UNDER DISASTER PREPAREDNESS OF TRANSMISSION INFRASTRUCTURE		
TITLE	STRUCTURAL DRAWING OF UNDERGROUND BUNKER		
SCALE	AS SHOWN	DATE	20/08/2018
BY	C/ENG/UG/BUNKER/STR/01	REV	2-0



FOR TENDER/ CONSTRUCTION			
POWER GRID CORPORATION OF INDIA LIMITED (A Government of India Enterprise)			
PROJECT	CONSTRUCTION OF BUNKER IN CRITICAL SWITCHYARD/ ESTABLISHMENT UNDER DISASTER PREPAREDNESS OF TRANSMISSION INFRASTRUCTURE		
TITLE	STRUCTURAL DRAWING OF UNDERGROUND BUNKER		
SCALE	AS SHOWN	DATE	20/08/2018
BY	C/ENG/UG/BUNKER/STR/01	REV	2-0

2.4. METHODOLOGY

The preparation of the BOQ and BBS followed a systematic, drawing-to-estimate workflow:

- **Step 1 :** Studying architectural and structural drawings to understand the geometry, levels, and reinforcement layout of every structural member (footings, walls, columns, beams, slabs, staircase).
- **Step 2 :** Preparing a detailed quantity take-off (calculation sheet) for each civil item applicable as per Approved Drawing.
- **Step 3 :** Preparing the Bar Bending Schedule (BBS) member-wise, computing the cutting length of every bar (accounting for bends, hooks, and development length) and the total steel weight required, segregated dia-wise (8 mm, 10 mm, 12 mm, 16 mm, 20 mm).
- **Step 4 :** Applying applicable DSR-2023 (Delhi Schedule of Rates) item codes and rates including applicable cost index for this area to each measured quantity to arrive at the Bill of Quantities, and adding non-schedule items (sanitary and finishing fixtures) based on rates from recently awarded LOAs (Letters of Award) or market rates.
- **Step 5 :** Cross-checking computed quantities against the drawings and getting the BBS and BOQ verified/checked by the site engineer.

QUANTITY ESTIMATION

[illegible]

	Extra for every additional lift of 1.5 m or part thereof in.: All kinds of Soil		Cum	1							
5	PCC										
	Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level	4.1.3									
a	1:2:4 (1 Cement : 2 coarse sand (zone-III) : 4 graded stone aggregate 20 mm nominal size).										
	Septic Tank		Cum	1		3.14	0.56	0.23		0.40	
	Total =									0.40	
b	1:3:6 (1 Cement : 3 coarse sand (zone-III) : 6 graded stone aggregate 20 mm nominal size)	4.1.5									
	Bunker (Sunken Portion near Toilet)		Cum	1		6.59	3.21	0.075		1.59	5.24
	Bunker (Remaining Portion)		Cum	1		10.53	3.34	0.075		2.64	2.637765
	Bunker (Footing Portion near to entry)		Cum	1		6.59	3	0.075		1.48	15.24
	Footing Trench along the length at remaining portion		Cum	2		10.53	1.7	0.075		2.69	
	Septic Tank		Cum	1		2.26	2.26	0.300		1.53	
	Total =									9.92	
6	RCC										
	Providing and laying in position specified grade of reinforced cement concrete, excluding the cost of centering, shuttering, finishing and reinforcement - All work up to plinth level :	5.1									
a	1:1.5:3 (1 cement : 1.5 coarse sand (zone-III): 3 graded stone aggregate 20 mm nominal size)	5.1.2	cum								
	RCC Footings										
	Bunker (Sunken Portion near Toilet)- Raft		Cum	1		6.44	3.06	0.550		10.84	
	Bunker (Footing Portion near to entry)- Raft		Cum	1		6.44	2.85	0.550		10.09	
	Bunker (Remaining Portion)-Raft		Cum	1		10.53	3.34	0.300		10.55	
	Bunker (Remaining Portion) near the Toilet Interior Wall		Cum	1		3.34	0.6	0.450		0.90	
	Footing Trench along length		Cum	2		10.53	1.55	0.550		17.95	
	RCC Straight Walls Below Slab Height										
	Bunker Walls of Toilet - Short Walls		Cum	2		1.21	0.35	3.450		2.92	
	Bunker Walls of Toilet - Long Walls		Cum	2		5.24	0.35	3.450		12.65	
	Bunker Walls of Toilet - (Partition Walls)		Cum	1		1.21	0.3	3.450		1.25	
	Deduction for Doors		Cum	-2		0.8	0.3	2.100		-1.01	

	Bunker Long Walls (Remaining Portion)		Cum	2		13.33	0.35	3.000		27.99	
	Bunker Short Walls (Remaining Portion)		Cum	1		4.54	0.35	3.000		4.77	
	Total =									98.92	
6	RCC ABOVE SLAB HEIGHT										
	Reinforced cement concrete work in walls (any thickness), including attached pilasters, buttresses, plinth and string courses, fillets, columns, pillars, piers, abutments, posts and struts etc.above plinth level up to floor five level, excluding cost of centering,shuttering, finishing and reinforcement :	5.2	cum								
a	1:1.5:3 (1 cement : 1.5 coarse sand (zone-III) : 3 graded stone aggregate 20 mm nominal size)	5.2.2									
	Elevation B										
	Portion A Rectangular Wall		Cum	1		5.24	0.35	1.250		2.29	
	Portion B Trapezium Wall		Cum	1		2.69	0.35	1.300		1.22	
	Portion C (Chajja)		Cum	1		0.36		0.35		0.126	
	Elevation D Side										
	Portion A Rectangular Wall		Cum	1		1.85	0.35	0.950		0.62	
	Portion B Triangular Wall		Cum	1		0.57		0.350		0.20	
	Septic Tank		Cum	1		3.76	3.76	3.83		54.08	
	Elevation C Side										
	Portion A Rectangular Wall		Cum	1		0.95	0.35	3.480		1.16	
	Portion B Trapezium Wall		Cum	1		2.69	0.35	1.300		1.22	
	Portion C (Chajja)		Cum	1		0.36		0.35		0.126	
	Column (Typical)		Cum	2		3	0.3	0.300		0.54	
	Total =									61.58	
8	RCC SLAB										
	Reinforced cement concrete work in beams, suspended floors, roofs having slope up to 15° landings, balconies, shelves, chajjas, lintels, bands, plain window sills, staircases and spiral stair cases up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement, with 1:1.5:3 (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20 mm nominal size).	5.3									
	Slab of Bunker at +0.6 metre level (Part-1)		Cum	1		12.79	6.44	0.300		24.71	
	Slab of Bunker at +0.6 metre level (Part-2)		Cum	1		4.38	1.7	0.300		2.23	
	Slab of Mumty at +1.55 metre level (Flat)-backside		Cum	1		1.76	1.85	0.300		0.98	
	Slab of Mumty at +1.55 metre level (Inclined)-backside		Cum	1		1.73	1.76	0.300		0.91	

	Slab of Mumty at +3.15 metre level (Flat)-Frontside		Cum	1	2.5	1.65	0.300	1.24	
	Slab of Mumty at +3.15 metre level (Inclined)-Frontside		Cum	1	2.25	1.65	0.300	1.11	
	Tie Beam-1 at 0.6 metre level		Cum	3	4.54	0.3	0.450	1.84	
	Tie Beam-2 at 0.6 metre level		Cum	1	3.13	0.3	0.600	0.56	
	Mumty Beam (At all levels)		Cum	1	5.21	0.3	0.150	0.23	
	Stringer Beam (At all levels)		Cum	2	5.15	0.3	0.300	0.93	
	Stair Slab(landing1+flight1+landing2)		Cum	1	5.15	1.3	0.15	1.00	
	Stair Slab(flight2)		Cum	1	1.77	1.11	0.15	0.29	
	Total =							36.05	
9	CENTERING AND SHUTTERING								
	Centering and shuttering including strutting, propping etc. and removal of form for all heights								
a	Foundations, footings, bases of columns, etc. for mass concrete (Item code no: 5.9.1)	5.9.1	sqm						
	Bunker (Sunken Portion near Toilet)- Raft		sqm	1	19		0.550	10.45	
	Bunker (Footing Portion near to entry)- Raft-OUTER EDGE		sqm	1	33.2		0.550	18.26	
	Bunker (Footing trench along length)		sqm	1	24.4		0.250	6.10	
	Bunker (Remaining Portion)-Raft		sqm	1	3.34		0.150	0.50	
	Septic Tank		sqm	1	9.04	1	0.300	2.71	
	Total =							38.02	
b	Walls (any thickness) including attached pilasters, buttresses,plinth and string courses etc. (Item code no: 5.9.2)	5.9.2	sqm						
	Exterior Walls upto 3 m height			1	32		3.000	96.00	
	Interior Walls upto 3 m height			1	35.14		3.000	105.42	
	Exterior Walls upto 3.45 m height			1	8.96		3.450	30.91	
	Interior Walls upto 3.45 m height			2	6.66		3.450	45.95	
	Deductions							0.00	
	Toilet Doors			-2	0.8		2.100	-3.36	
	Walls near to Mumty (upto 1.55m from ground along short wall)		sqm	1	2.95		0.950	2.80	
	Walls near to Mumty (around slope @1.55m from GL along short wall)		sqm	1	1.60		0.950	2.13	
	Walls near to Mumty (around slope @1.55m from GL along short wall)		sqm	1	1.60		0.650	1.65	
	Walls near to Mumty (upto 1.55m from ground along length)		sqm	1	3.17		0.950	3.01	
	Walls near to Mumty (upto 1.55m from ground along length)		sqm	1	6.96		0.950	6.61	
	Walls near to Mumty (upto 1.55m from ground along length)		sqm	1	6.96		0.650	4.52	
	Walls near to Mumty(@3.15m from ground along length)		sqm	2	3.8		1.600	12.16	
	Walls near to Mumty (around slope @3.15m from GL along short wall)		sqm	2	0.79		1.600	3.37	
	Typical wall near Mumty		sqm	1	0.65		2.550	1.66	
	Total =							312.83	
c	Suspended floors, roofs, landings, balconies and accessplatform.(Item code no: 5.9.3)	5.9.3	sqm						
	Slab of Bunker at +0.6 metre level (Part-1)		sqm	1	50.51		0.300	15.15	
			sqm	1	11.24		5.74	64.52	
	Deductions : Partition Wall of Toilet		sqm	-1	1.21		0.3	-0.36	
	Slab of Bunker at +0.6 metre level (Part-2)		sqm	1	4.38		1.700	7.45	
	Deductions :	TB2	sqm	1	-4.38		0.300	-1.31	
	Slab of Mumty at +1.55 metre level (Flat)-backside		sqm	1	1.41		1.200	1.69	
	Slab of Mumty at +1.55 metre level (Inclined)-backside		sqm	1	1.73		1.110	1.92	
	Slab of Mumty at +3.15 metre level (Flat)-Frontside		sqm	1	2.15		1.000	2.15	

	Slab of Mumty at +3.15 metre level (Inclined)-Frontside		sqm	1		2.25		1.000		2.25	
	Tie Beam-1 at 0.6 metre level		sqm	3		4.54		0.600		8.17	
	Tie Beam-2 at 0.6 metre level		sqm	1		3.13		1.200		3.76	
	Mumty Beam (At all levels)		sqm	1		5.21		0.600		3.13	
	Stringer Beam (At all levels)		sqm	2		5.15		0.600		12.35	
	Total =									120.86	
d	Plain cement concrete for mass concrete (Item code no: 4.1.5)	5.9.1	sqm								
	Bunker (Sunken Portion near Toilet)- Raft		sqm	1		19.3		0.075		1.45	
	Bunker (Footing Portion near to entry)- Raft		sqm	1		15.63		0.075		1.17	
	Bunker (Footing trench along length)		sqm	2		20.91		0.075		3.14	
	Septic Tank		sqm	1		9.04	1	0.300		2.71	
	Total =									8.47	
e	Columns, Pillars, Piers, Abutments, Posts and Struts.(Item code no: 5.9.6)	5.9.6	sqm								
			sqm	2		1.2	1	3.800		9.12	
	Total=									9.12	
f	Stairs, (excluding landings) except spiral-staircases (Item code no: 5.9.7)	5.9.7	sqm								
	Stair Slab(landing1+flight1+landing2)		sqm	1		5.15		1.000		5.15	
	Stair Slab(flight2)		sqm	1		1.77		1.410		2.49	
	Total=									7.64	
10	PLASTERING										
a	6 mm cement plaster 1:3 (1 cement : 3 fine sand)										
	Interior plastering of walls-Bunker portion	13.16.1	sqm		2	13.03		3		78.18	
	Interior plastering of walls-Toilet Portion		sqm		2	4.54		3		27.24	
	Interior plastering of walls-Toilet Portion		sqm		4	1.21		3		14.52	
	Interior plastering of walls-Toilet Portion		sqm		4	2.12		3		25.44	
	Elevation-D Exterior Wall rectangular portion		sqm		1	1.65		0.950		1.57	
	Elevation-D Interior Wall rectangular portion		sqm		1	1.3		0.950		1.24	
	Elevation-D Exterior Wall Triangular Portion		sqm		2	0.80		0.950		1.52	
	Elevation-D Interior Wall Triangular Portion		sqm		2	0.80		0.650		1.04	
	Elevation-B Exterior Wall rectangular portion		sqm		1	1.76		0.950		1.67	
	Elevation-B Interior Wall rectangular portion		sqm		1	1.41		0.950		1.34	
	Elevation-B Exterior Wall rectangular portion		sqm		2	3.48		0.950		6.61	
	Elevation-B Interior Wall rectangular portion		sqm		2	3.13		0.950		5.95	
	Elevation-B Exterior Wall rectangular portion		sqm		2	1.9		1.300		4.94	
	Elevation-B Interior Wall rectangular portion		sqm		2	1.9		1.300		4.94	
	Elevation-B Exterior & Interior Wall Triangular portion		sqm		4	0.79		1.300		4.11	
	Typical wall near Mumty-Exterior		sqm		1	0.35		2.550		0.89	
	Typical wall near Mumty-Exterior		sqm		1	0.3		2.250		0.675	
	(-) Door Deduction		sqm		-2	0.8		2.1		-3.36	
	Bottom of slab/ceiling-Interior(Centre part)		sqm		1	11.73		3.43		40.23	
	Bottom of slab/ceiling-Interior(Toilet portion)		sqm		2	1.21		2.12		5.13	
	Bottom of slab/ceiling-Interior(Entry portion)		sqm		1	5.21		1.3		6.77	
	Bottom of slab/ceiling-Interior(Slope2)		sqm		1	2.93		1.11		3.25	
	Projections of slab-Exterior		sqm		1	33.87		0.6		20.32	
	Column		sqm		2	1.2		3.800		9.12	
	TB 1		sqm		6	4.54		0.15		4.09	
	TB 2		sqm		2	3.43		0.3		2.06	
	MB		sqm		1	5.91		0.15		0.89	
	SB		sqm		2	5.84		0.9		10.51	
	Edges of slabs				1	50.51		0.300		15.15	

	conforming to CPWD specification & direction of Engineer-in-Charge. Section of steel reinforcement and cross sections of uPVC profiles to be as per design approved by Engineer-in-Charge. Wire mesh / Glazing of plain/ toughened/ laminated/ double glass unit with / without high performance coatings as per design requirements and conforming to IS: 3548 & IS: 16231 shall be paid separately. Note:- Structural design proof checked from a Government Engineering Institute, to be provided by the manufacturer for : (i) Sites with basic wind speed > 45 m/sec as per IS 875 - Part 3 (ii) Sites with structure height									
	Casement door single panel with zinc alloy (zamak) 3D hinges, multi-point locking system with cylinder, zinc alloy (zamak) powder coated handles on both sides.	9.147.F1								
	Using R3 series with frame (55 mm & above) x (50mm & above) & sash (55 mm & above) x (85mm & above) (Height upto 2.5 metre).	9.147.F1.1								
			sqm	1		0.8	2			1.60
			sqm	1		0.8	2			1.6
	Total =									3.20
16	ENTRY DOOR									
	Providing and fixing 1 mm thick M.S. sheet door with frame of 40x40x6mm angle iron and 3 mm M.S. gusset plates at the junctions and corners, all necessary fittings complete, including applying a priming coat of approved steel primer	10.5.1	sqm							
			sqm	1		1.2	2.65			3.18
	Total =									3.18
17	CEMENT CONCRETE FLOORING									
	Cement concrete flooring 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate) finished with a floating coat of neat cement, including cement slurry, but excluding the cost of nosing of steps etc. complete.									
a	a. 40 mm thick with 20 mm nominal size stone aggregate (Item code no: 11.3.1)	11.3.1								
	Bunker		sqm	1		11.73	4.54			53.25
	Staircase and Landing		sqm	1		4.54	1.3			5.902
	Total =									59.16
18	FLOORING IN TOILET									
	Providing and laying Vitrified tiles in floor in different sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS:15622, of approved brand & manufacturer, in all colours and shade, laid on 20 mm thick cement mortar 1:4 (1 cement: 4 coarse sand) jointing with grey cement slurry @3.3 kg/sqm including grouting the joints with white cement and matching pigments etc. The tiles must be cut	11.41A.3								

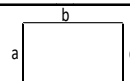
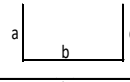
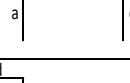
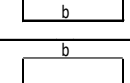
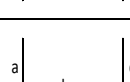
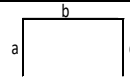
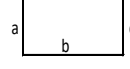
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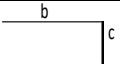
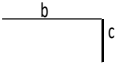
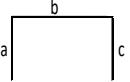
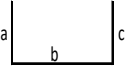
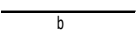
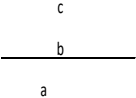
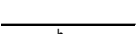
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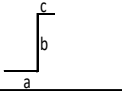
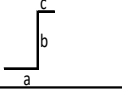
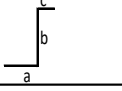
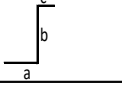
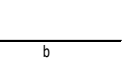
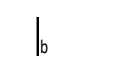
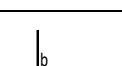
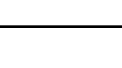
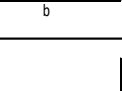
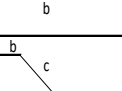
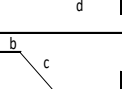
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BAR BENDING SCHEDULE(BBS)

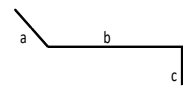
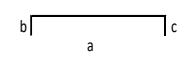
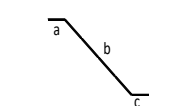
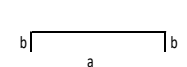
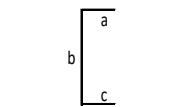
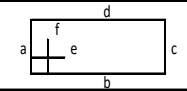
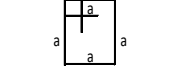
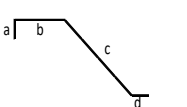
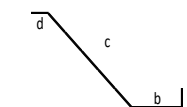
BAR BENDING SCHEDULE (BBS)

Name of the Work:																							
Sr. No.	Particular	Bar Mark	Dia of bar (mm)	Spa.@c/c (mm)	Shape and Size	Break Up (mm)							Cutting Length (mm)	No. of Identical Items	No. of Bar	Total Length (Mtr.)	Dia wise cutting length(Mtr.)						
						a	b	c	d	e	f	g					6	8	10	12	16	20	
	Sunken portion for Toilet																						
1	Footing F1(6440*3060MM)	Along Longer Side(Top)	12	150		450	6340	450					7200	1	21	149.280				149.280			
2	Footing F1(6440*3060MM)	Along Longer Side(Bottom)	12	150		450	6340	450					7200	1	21	149.280				149.280			
3	Footing F1(6440*3060MM)	Along Shorter Side(Top)	12	150		450	2960	450					3820	1	43	165.280				165.280			
4	Footing F1(6440*3060MM)	Along Shorter Side(Bottom)	12	150		900	2960	450	600				4870	1	43	210.710				210.710			
5	Footing F1(2850*6440MM)	Along Longer Side(Top)	12	150		450	6340	450					7200	1	19	139.200				139.200			
6	Footing F1(2850*6440MM)	Along Longer Side(Bottom)	12	150		450	6340	450					7200	1	19	139.200				139.200			
7	Footing F1(2850*6440MM)	Along Shorter Side(Top)	12	150		450	2750	450					3610	1	43	156.190				156.190			
8	Footing F1(2850*6440MM)	Along Shorter Side(Top)	12	150		450	2750	450					3610	1	43	156.190				156.190			

9	Footing F1(10530*1550MM)	Along Longer Side(Top)	12	150			11080	926					11966	2	11	255.270				255.270			
10	Footing F1(10530*1550MM)	Along Longer Side(Bottom)	12	150			11080	476					11516	2	11	245.670				245.670			
11	Footing F1(10530*1550MM)	Along Shorter Side(Top)	12	150		450	1450	450					2310	2	75	345.880				345.880			
12	Footing F1(10530*1550MM)	Along Shorter Side(Bottom)	12	150		450	1450	450					2310	2	75	345.880				345.880			
13	Walls in Longer Direction (5240*4000 MM)	Bar at Outer Face in Vertical Direction	16	175		450	4177	300					4927	1	31	151.050				151.050			
14	Walls in Longer Direction (5240*4000 MM)	Bar at Inner Face in Vertical Direction	12	175		450	4177	300					4927	1	27	132.750				132.750			
15	Interior Wall in Longer Direction (5240*4000 MM)	Bar at both face in Vertical Direction	10	175		450	4177	250					4877	2	27	262.800			262.800				
16	Both Walls for Toilet	Distribution Bar	10	175			5190						5190	4	21	430.029			430.029				
17	Partition wall of Toilet	Bar at both face in Vertical Direction	10	175		450	3877	250					4577	2	8	72.450			72.450				
18	Partition wall of Toilet	Distribution Bar	10	175			1210						1210	2	21	50.129			50.129				

19	Outer WallsParallel to Partition Wall (2460*4300 MM)	Bar at Outer Face in Vertical Direction	16	175		450	4177	300					4927	2	15	146.970					146.970		
20	Outer WallsParallel to Partition Wall (2460*4300 MM)	Bar at Inner Face in Vertical Direction	12	175		450	4177	300					4927	2	13	128.670				128.670			
21	Outer Walls along longer direction (12780*3850 MM)	Bar at Outer Face in Vertical Direction	16	175		450	3727	300					4477	2	74	660.290					660.290		
22	Outer Walls along longer direction (12780*3850 MM)	Bar at Inner Face in Vertical Direction	12	175		450	3727	300					4477	2	72	644.940				644.940			
23	Outer Walls along longer direction (12780*3850 MM)	Distribution Bar	10	175			12730						12730	4	18	923.834			923.834				
24	ELE-D Wall below the Mumty Slab	Bar at Outer Face in Vertical Direction	16	175			901						901	2	13	22.910					22.910		
25	ELE-D Wall below the Mumty Slab	Bar at Inner Face in Vertical Direction	12	175			901						901	2	11	19.310				19.310			
26	ELE-D Wall below the Mumty Slab	Distribution Bar	10	175			2025						2025	2	6	26.036			26.036				
27	Top Flat Slab near the Entry gate(14190*4080 MM)	Bottom & Top Bar	12	150		450	14140	250					14840	2	27	797.400				797.400			
28	Top Flat Slab back side of Entry gate	Top Bar	12	150		250	1850	2030	13040	250			17420	1	16	273.950				273.950			
29	Top Flat Slab back side of Entry gate	Bottom Bar	12	150		250	2050	2030	12790	250			17370	1	16	273.160				273.160			

30	Top Flat Slab back side of Entry gate	Top/Bottom Bar	12	150		250	12740	250					13240	2	4	105.830				105.830			
31	TB 1 (300 x 450 mm)	Bottom Bar	25			400	5190	400					5990	3	3	53.910							54
32	TB 1 (300 x 450 mm)	Top Bar	20			400	5190	400					5990	3	3	53.910					53.910		
33	TB 1 (300 x 450 mm)	STPS	10	150		250	400	250	400	100	100		1500	3	31	140.700			140.700				
34	TB2 (300 x 600mm)	Bottom Bar	25			550	3730	550					4830	1	3	14.490							14
35	TB2 (300 x 600mm)	Top Bar	20			550	3730	550					4830	1	3	14.490					14.490		
36	TB2 (300 x 600mm)	STPS	10	150		250	550	250	550	100	100		1800	1	22	39.360			39.360				
37	STAIRCASE (Landing1+Flight1+Landing2)	Top Bar	12	150		100	1255	3336	300				4991	1	9	43.251				43.251			
38	STAIRCASE (Landing1+Flight1+Landing2)	Bottom Bar	12	150		100	1435	3336	300				5171	1	9	44.811				44.811			
39	STAIRCASE (Landing1+Flight1+Landing2)	Top Edge bars	12	150		100	1435	300					1835	1	9	15.906				15.906			

40	STAIRCASE (Landing1+ Flight1+Landing2)	Bottom Edge bars	12	150		300	1615	100						2015	1	9	17.466				17.466			
41	STAIRCASE (Landing1+ Flight1+Landing2)	Distribution Bar at Top & Bottom	12	150		1250	144	144						1538	2	41	126.640				126.640			
42	STAIRCASE (Flight2)	Top Bar	12	150		300	1860	300						2460	2	8	41.333				41.333			
43	STAIRCASE (Flight2)	Distribution Bar	12	150		1060	100	100						1060	2	12	26.292				26.292			
44	RCC Column 1		12			250	3746	600						4596	2	8	73.540				73.540			
45	RCC Column 1	STPS	8	190		220	220	220	220	80	80			1040	2	14	29.558		29.558					
46	RCC Column 1	STPS	8	190		156	156	156	156	80	80			782	2	14	22.232		22.232					
47	MB (300*450mm)	Top Bars	20			400	1875	2851	600					5726	2	3	34.350						34.350	
48	MB (300*450 mm)	Bottom Bar	25			400	1735	2851	600					5586	2	3	33.510							34

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BILL OF QUANTITIES

BILL OF QUANTITY (BOQ)								
Name of Work: Construction of Under Ground Bunker at 400/132 KV POWERGRID Silchar Sub-Station.								
Sl.	DSR-2023	Item Description	Unit	Quantity	DSR Rate (Rs.)	Unit Rate Including CI @ 44.51%	DSR Rate Excluding GST (Rs.)	Amount (Rs.)
<u>Civil Portion</u>								
Schedule items - (Items available in DSR-2023)								
1	1.1.2	By mechanical transport including loading, unloading and stacking (upto 2 km lead).	cum	847.62	245.82	355.21	301.03	255159.41
2	2.6.1	Earth work in excavation by mechanical means (Hydraulic excavator)/manual means over areas (exceeding 30 cm in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5 m, as directed by Engineer-in-charge. : All Kinds of soil	cum	556.86	177.5	256.49	217.36	121039.38
3	2.25	Filling available excavated earth (excluding rock) in trenches, plinth, sides of foundations etc. in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 m and lift upto 1.5 m.	cum	290.76	196	283.22	240.02	69788.17
4	2.26.1	Extra for every additional lift of 1.5 m or part thereof in.: All kinds of Soil	cum	314.36	126.80	183.23	155.28	48813.12
5	4.1.3	Providing and laying in position cement concrete of specified grade excluding the cost of centring and shuttering : All works upto plinth level-1:2:4 (1 Cement : 2 coarse sand (zone-III) : 4 graded stone aggregate 20 mm nominal size).	cum	0.40	7878.5	11384.43	9647.82	3901.89
6	4.1.5	Providing and laying in position cement concrete of specified grade excluding the cost of centring and shuttering : All works upto plinth level-1:3:6 (1 Cement : 3 coarse sand : 6 graded stone aggregate 20 mm nominal size).	cum	9.92	7294.7	10540.84	8932.92	88654.65
7	5.1.2	Providing and laying in position specified grade of reinforced cement concrete excluding the cost of centring, shuttering, finishing and reinforcement- All works upto Plinth level : 1:1.5:3 (1 Cement : 1.5 Coarse sand : 3 graded stone aggregate 20 mm nominal size)	cum	98.92	9045.75	13071.11	11077.21	1095766.81
8	5.2.2	Reinforced cement concrete work in walls (any thickness), including attached pilasters, buttresses, plinth and string courses, fillets, columns, pillars, piers, abutments, posts and struts etc. up to floor five level, excluding cost of centering, shuttering, finishing and reinforcement : 1:1.5:3 (1 Cement: 1.5 Coarse sand: 3 Graded stone aggregate 20 mm nominal size)	cum	7.50	10852.95	15682.51	13290.26	99731.77
9	5.3	Reinforced cement concrete work in beams, suspended floors, roofs having slope up to 15° landings, balconies, shelves, chajjas, lintels, bands, plain window sills, staircases and spiral stair cases up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement, with 1:1.5:3 (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20 mm nominal size).	cum	36.05	11505.50	16625.45	14089.36	507865.09

10	5.9.1	Centring and shuttering including strutting, propping etc. and removal of formwork for : Foundations, footings, bases of columns etc. for mass concrete.	sqm	38.02	392.15	566.66	480.22	18259.41
11	5.9.2	Centring and shuttering including strutting, propping etc. and removal of formwork for : Walls (any thickness) including attached pilasters, buttresses	sqm	312.83	842.5	1217.41	1031.7	322751.46
12	5.9.3	Suspended floors & Slabs, roofs, landings, balconies and access platform	sqm	120.86	927.25	1339.88	1135.49	137232.38
13	5.9.6	Centring and shuttering including strutting, propping etc. and removal of formwork for : Columns, pillars, piers, abutments, posts & struts	sqm	9.12	961.3	1389.08	1177.19	10735.97
14	5.9.7	Centring and shuttering including strutting, propping etc. and removal of formwork for : Stairs (excluding landings) except spiral staircases	sqm	7.64	764.9	1105.28	936.68	7152.13
15	5.22.6	Reinforcement for R.C.C. work including straightening, cutting, bending, binding and placing in position complete : Thermo-Mechanically Treated bars	kg	8605.30	107.85	155.84	132.07	1136502.53
16	6.1.2	Brick work with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in foundation and plinth in::Cement mortar 1:6 (1 cement : 6 coarse sand)	cum	4.31	7132.25	10306.1	8733.98	37619.01
17	13.16.1	6 mm cement plaster to RCC work of mix 1 : 3 (1 Cement : 3 Fine sand)	sqm	308.32	300.45	434.15	367.92	113437.77
18	13.47.1	Finishing walls with Premium Acrylic Smooth exterior paint with Silicone additives of required shade :New work (Two or more coats applied @ 1.43 ltr/10 sqm over and including priming coat of exterior primer applied @ 0.90 litre/10 sqm)	sqm	67.70	171.10	247.24	209.53	14184.34
19	13.83.2	Wall painting with premium acrylic emulsion paint of interior grade, having VOC (Volatile Organic Compound) content less than 50 grams/ litre of approved brand and manufacture, including applying additional coats wherever required to achieve even shade and colour :Two coats	sqm	240.63	142.80	206.35	174.87	42078.24
20	13.85.3	Applying priming coats with primer of approved brand and manufacture, having low VOC (Volatile Organic Compound) content : With water thinnable cement primer on wall surface having VOC content less than 50 grams/litre	sqm	240.63	142.80	206.35	174.87	42078.24
21	8.31	Providing and fixing 1st quality ceramic glazed wall tiles conforming to IS: 15622 (thickness to be specified by the manufacturer), of approved make, in all colours, shades except burgundy, bottle green, black of any size as approved by Engineer-in-Charge, in skirting, risers of steps and dados, over 12 mm thick bed of cement mortar 1:3 (1 cement : 3 coarse sand) and jointing with grey cement slurry @ 3.3kg per sqm, including pointing in white cement mixed with pigment of matching shade complete	sqm	24.53	1267.95	1832.19	1552.7	38084.63
22	9.147.F1.1	Providing and fixing factory made UPVC glazed/wire mesh windows/doors comprising of lead free uPVC multi-chambered frame, sash and mullion/coupler (where ever required) extruded profiles having minimum wall thickness of 1.70 mm for Series R1 and R2 profiles and 2.10 mm for Series R3 and R4 profiles conforming to EN: 12608 in any shape, colour and design duly reinforced with galvanized mild steel section made of required shape & size as per CPWD Specification, uPVC extruded glazing beads, interlocks and Inline sash adaptor (where ever required) of appropriate dimension, EPDM gasket, hardware, SS 304 grade fasteners of minimum 8 mm dia with countersunk head, comprising of matching polyamide PA6 grade sleeve for fixing frame to finished wall as per IS 1367 : Part 1 to 14, plastic packers, plastic caps and necessary stainless steel screws etc. Profile of frame, sash & mullion (if required) shall be mitred cut and fusion welded/mechanically jointed duly sealed at all corners, including drilling of holes for fixing hardware and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealant over backer rod of approved size and quality, all complete as per approved drawing conforming to CPWD specification & direction of Engineer-in-Charge. Section of steel reinforcement and cross sections of uPVC profiles to be as per design approved by Engineer-in-Charge. Wire mesh / Glazing of plain/ toughened/ laminated/ double glass unit with / without high performance coatings as per design requirements and conforming to IS: 3548 & IS: 16231 shall be paid separately. Note:- Structural design proof checked from a Government Engineering Institute, to be provided by the manufacturer for : (i) Sites with basic wind speed > 45 m/sec as per IS	Sqm	3.20	12496.05	18056.79	15302.36	48967.55

		875 – Part 3 (ii) Sites with structure height more than 20m for all wind speeds : Casement door single panel with zinc alloy (zamak) 3D hinges, multi-point locking system with cylinder, zinc alloy (zamak) powder coated handles on both sides--Using R3 series with frame (55 mm & above) x (50mm & above) & sash (55 mm & above) x (85mm & above) (Height upto 2.5 metre)						
23	10.5.1	Providing and fixing 1 mm thick M.S. sheet door with frame of 40x40x6mm angle iron and 3 mm M.S. gusset plates at the junctions and corners, all necessary fittings complete, including applying a priming coat of approved steel primer	sqm	3.18	5804.35	8387.29	7107.87	22603.03
24	11.3.1	Cement concrete flooring 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate) finished with a floating coat of neat cement, including cement slurry, but excluding the cost of nosing of steps etc. complete : 40mm thick with 20mm nominal size stone aggregate	sqm	59.16	614.2	887.52	752.14	44493.74
25	11.41A.3.1	Providing and laying Vitrified tiles in floor in different sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS:15622, of approved brand & manufacturer, in all colours and shade, laid on 20 mm thick cement mortar 1:4 (1 cement: 4 coarse sand) jointing with grey cement slurry @3.3 kg/sqm including grouting the joints with white cement and matching pigments etc. The tiles must be cut with the zero chipping diamond cutter only . Laying of tiles will be done with the notch trowel, plier, wedge, clips of required thickness, leveling system and rubber mallet for placing the tiles gently and easily : Glazed Vitrified tiles Matt/Antiskid finish of size--Size of Tile 600 x 600 mm	sqm	5.09	1464.85	2116.71	1793.82	9126.96
26	18.7.3	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply and all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge : Internal work - Exposed on wall--25 mm nominal outer dia Pipes	each	30.00	401.55	580.24	491.73	14751.90
27	18.8.2	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings and fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge : Concealed work, including cutting chases and making good the walls etc--20 mm nominal outer dia Pipes	each	30.00	537.6	776.83	658.33	19749.90
28	12.41.1	Providing and fixing on wall face unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion, (i) Single socketed pipes: 75 mm dia	Metre	10.00	248.80	359.52	304.68	3046.80
29	12.41.2	Providing and fixing on wall face unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion, (i) Single socketed pipes: 110 mm dia	Metre	20.00	377.40	545.34	462.15	9243.00
30	12.42.1.1	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion: Coupler 75 mm dia	Each	8.00	91.70	132.51	112.3	898.40
31	12.42.1.2	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion: Coupler 110 mm dia	Each	6.00	136.15	196.74	166.73	1000.38

32	12.42.5.1	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion: Bend 87.5--75 mm bend	Each	6.00	105.90	153.03	129.69	778.14
33	12.42.5.2	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion: Bend 87.5--100 mm bend	Each	6.00	150.35	217.26	184.12	1104.72
34	12.42.3.1	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion: Single tee with door 75x75x75 mm	Each	2.00	164.20	237.27	201.08	402.16
35	12.42.3.2	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion: Single tee with door 75x75x75 mm dia 110x110x100 mm	Each	2.00	234.15	338.35	286.74	573.48
Total Amount (Exclusive of GST):								₹ 43,70,529.47
A= Sub-Total Cost for Scheduled items (Inclusive of GST @18%):								₹ 51,57,224.78

Non-Schedule items - (Rates taken from recently awarded LOA)

1	LOA Rates	Providing and fixing UPVC SWR Pipe fittings for Soil and waste discharge system conforming to IS : 14735, including jointing with seal ring and solvent : Vent Cowl 75 mm dia	Each	1.00			200.00	200.00
2	LOA Rates	Providing and fixing UPVC SWR Pipe fittings for Soil and waste discharge system conforming to IS : 14735, including jointing with seal ring and solvent : Vent Cowl 110 mm dia	Each	2.00			200.00	400.00
3	LOA Rates	Providing and fixing UPVC SWR Pipe fittings for Soil and waste discharge system conforming to IS : 14735, including jointing with seal ring and solvent : Floor Trap 110x75 mm dia	Each	4.00			300.00	1200.00
4	LOA Rates	Providing and fixing UPVC SWR Pipe fittings for Soil and waste discharge system conforming to IS : 14735, including jointing with seal ring and solvent : Floor Trap 110x110 mm dia	Each	2.00			300.00	600.00
5	LOA Rates	Providing and fixing Health Faucet with 1.2 m flexible tube and wall hook Jaquar make Item code ALD-573 all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1600.00	3200.00
6	LOA Rates	Providing and fixing click clack Waste Coupling 32 mm size full thread Jaquar make item code ALD-727 all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1000.00	2000.00
7	LOA Rates	Providing and fixing Waste Coupling 32 mm Half thread Jaquar make item code ALD-709 all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1000.00	2000.00
8	LOA Rates	Providing and fixing Bottle Trap (with internal partition) 32mm size 250 mm & 190 mm long wall connection pipes & wall flange Jaquar make item code ALD-769L250x190 all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1800.00	3600.00

9	LOA Rates	Providing and fixing Single Piece-WC with UF Soft Close Seat Cover, Hinges, Dual Flush Cistern Fitting, fixing accessories and accessories set, cistern fittings, S Trap-300 mm size (390x710x730) mm Jaquar make item code VGS-WHT-81853S300UF all complete. or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			23200.00	46400.00
10	LOA Rates	Providing and fixing Angular stop cock with wall flange Jaquar make item code OPP-15053PM all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	12.00			950.00	11400.00
11	LOA Rates	Providing and fixing 2 Way Bib Cock with wall flange Jaquar make item code OPP-15041PM all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			2000.00	4000.00
12	LOA Rates	Providing and fixing exposed part kit of concealed stop cock & flush cock with fitting sleeve, operating lever & adjustable wall flange with seal (compatible with ALD-083, ALD-089 & ALD-081) Jaquar make item code OPP-15083KPM all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1600.00	3200.00
13	LOA Rates	Providing and fixing Single Lever Tall Boy with 150mm Extension Body Fixed Spout without Popup Waste System with 600 mm long braided hoses Jaquar make item code OPP-15005BPM all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			5500.00	11000.00
14	LOA Rates	Providing and fixing Single Towel Rail 600mm Long, Stainless Steel Jaquar make item code AKP-35711PS all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			2400.00	4800.00
15	LOA Rates	Providing and fixing Soap Dish Holder Jaquar make item code AKP-35731P all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1050.00	2100.00
16	LOA Rates	Providing and fixing Reduced Body of concealed Stop Cock suitable for 15mm Pipe Line with Plastic Protection Cap (without exposed parts). Jaquar make item code ALD-069FT all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1100.00	2200.00
17	LOA Rates	Providing and fixing Storage Water Heater 15 litres Jaquar make item code Elena 5 Star Rating all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			10000.00	20000.00
18	LOA Rates	Providing and fixing Soap Dispenser Push Button Jaquar make item code SDR-WHT-DJ0010F all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			1200.00	2400.00
19	LOA Rates	Providing and fixing Hand Dryer Flash touch-free infrared hand dryer with 1150W rated power & carbon brush motor Material : 304 Stainless Steel and Finish: Stainless Steel Finish Jaquar make item code HDR-SSF-AK2807 all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			9200.00	18400.00
20	LOA Rates	Providing and fixing beveled edge mirror with minimum thickness of 8 mm of superior glass (of approved quality) complete fixed on wall with superior quality SS Studs and spacers of suitable dimensions as per approved drawing and as direction of Engineer-in-charge.	sqm	4.00			2000.00	8000.00

21	Market Quote	Providing and fixing wall hung corner basin Jaquar make item code JDS-WHT-25841 all complete or equivalent model in Hindware/ Parryware/ Cera/ Somany or equivalent.	each	2.00			6000.00	12000.00
22	Market Quote	Supplying and fixing C.I. cover with frame for manholes : 560 mm internal diameter C.I. cover (heavy duty) the weight of the cover to be not less than 108 kg	each	1.00			11000.00	11000.00
23	Market Quote	Supplying and fixing Aluminium Louvered ventilators with frames & blades made up of extruded aluminim & typical wall thickness of 1.58 mm.The blades are spaced 75 mm on centre & set at 45 degrees to horizontal. Flyproof SS Jali is to be attached to the louvered ventilator. All complete as per directions of EIC.	sqm	1.80			5600.00	10080.00
Total Amount (Exclusive of GST):								₹ 1,80,180.00
B= Sub-Total Cost for Non-Schedule items (Inclusive of GST @18%):								₹ 2,12,612.40
(A+B)= Total Amount for Civil Portion (Inclusive of GST):								₹ 53,69,837.18
Electrical Portion								
1	1.10.3	Wiring for light point/ fan point/ exhaust fan point/ call bell point with 1.5 sq.mm FRLS PVC insulated copper conductor single core cable in surface / recessed medium class PVC conduit, with modular switch, modular plate, suitable size GI. box and earthing the point with 1.5 sq.mm. FRLS PVC insulated copper conductor single core cable etc as required : Group C	Point	6.00			1467.00	8802.00
2	1.12	Wiring for light/Power plug with 2X4 sq.mm FRLS PVC insulated, copper conductor single core cable in surface/ recessed medium class PVC conduit alongwith 1 No. 4 sq.mm FRLS PVC insulated, copper conductor single core cable for loop earthing as required.	metre	20.00			334.00	6680.00
3	1.14.2	Wiring for circuit/submain wiring alongwith earth wire with the following sizes of FRLS PVC insulated copper conductor, single core cable in surface/recessed medium class PVC conduit as required : 2 x 2.5 Sq.mm + 1X2.5 sq.mm earth wire	metre	20.00			275.00	5500.00
4	1.31	Supplying and Fixing suitable size GI box with modular plate and cover in front on surface or in recess, including providing and fixing 3 pin 5/6 amps modular socket outlet and 5/6 modular switch, connection etc. as required.	each	4.00			477.00	1908.00
5	1.32	Supplying and Fixing suitable size GI box with modular plate and cover in front on surface or in recess, including providing and fixing 6 pin 5/6 & 15/16 amps modular socket outlet and 15/16 A modular switch, connection etc. as required.	each	2.00			586.00	1172.00
6	2.10.1	Supplying & fixing 5 to 32 Amps rating ,240/415 V, 10 kA, "C" Curve, miniature circuit breaker suitable for inductive load of following poles in the existing MCB DB complete with connections, testing and commissioning etc. as required : Single Pole	each	4.00			256.00	1024.00

7	LOA Value	Supplying, Installation, testing and commissioning of pre-wired LED fitting directly on ceiling/ wall, including connection with 1.5 sq. mm FRLS PVC insulated, copper conductor, single core cable and earthing etc. as required : Havells Trim clipon LED Panel - 6W or equivalent	each	6.00			1652.00	9912.00
8	LOA Value	Supplying, erecting, testing and commissioning of Exhaust fan for continuous duty in toilets /laboratories/bunkers etc. The required accessories for fixing of exhaust fan on wall / Frame shall be in scope of contractor and including Exhaust pipe with all the necessary clamping accessories to mount on ceiling upto the Exhaust point as per site requirement and providing necessary louvers/Shutters (as Applicable) : Havells Ventilair DXC 130 mm sweep White or equivalent	each	2.00			1888.00	3776.00
9	LOA Value	Supply & installation of high speed energy efficient Wall Fan with sweep of 450 mm of Havels or other reputed brand.	each	7.00			3000.00	21000.00
Total Amount (Exclusive of GST):								₹ 59,774.00
C= Sub-Total Cost for Electrical items (Inclusive of GST @18%):								₹ 70,533.32
(A+B+C) = Grand Total for Construction of Under Ground Bunker at 400/132 KV POWERGRID Silchar Sub-Station.								₹ 54,40,370.50

3. ABOUT THE PROJECT



-----PROJECT NAME-----

**Design and Construction of Reinforced Concrete Ramp with Anti-Skid Granite Finish
for Wheelchair Accessibility – Transit Camp & Control Room.**

As part of the accessibility upgrade works at the Transit Camp and Control Room premises, an RCC access ramp is proposed to be constructed outside the building, adjacent to the existing staircase. The ramp is designed as a wheelchair-accessible entry, allowing barrier-free movement for differently-abled personnel and visitors between the ground level and the building entrance, which is currently served only by stairs.

To accommodate the ramp within the available space, one side of the existing staircase will be demolished (cut back), and the ramp will be constructed in its place, running alongside the remaining flight. The ramp surface will be roughened/scratched to key the finish layers, with the balance of the work — base preparation, reinforcement, flooring, and railing — executed strictly as per the structural drawing.

3.1. SITE AND CONNECTIVITY



Figure 4 : Google Map image of PGCIL, Srikona

The geographic and logistical placement of the site plays a critical role in its construction supply chain (procurement of aggregate, cement, and steel).

- **Latitude & Longitude:** Control Room: 24°49'41.0"N 92°43'13.3"E
Transit Camp: 24°49'40.4"N 92°43'11.6"E
- **Location:** Power Grid 400/132 kV substation, Srikona, Assam
- **Connectivity:** Both the sites are connected directly to two lane National Highway (NH 37) that serves as a strategic transportation corridor linking Imphal (Manipur) to Silchar and Sribhumi in Assam, via Lalmati TV Tower Road at Indragarhgrant, ensuring all weather efficient connectivity. Its proximity to major transit corridors ensures the unhindered movement of heavy machinery, Transit Mixers, and Ready Mix Concrete (RMC) vehicles.

FACILITY	Wheelchair access ramp, Transit Camp & Control Room
LOCATION	Outside the building, adjoining the existing staircase (one side of stair demolished for construction)
TYPE	RCC ramp on PCC sub-base, with stone flooring and SS handrail

SLOPE	12.68° due to constraints of space
RAMP PLAN DIMENSION	≈ 2050 mm × 1200 mm
SURFACE TREATMENT	Anti-skid, linear strip pattern

3.2 STRUCTURAL DESCRIPTION & SPECIFICATIONS

-----SUB-BASE AND BASE COURSES-----

- I. 206.05 mm thick PCC (1:4:8) as the sub-base/foundation layer.
- II. 100 mm thick PCC (1:2:4) laid over the sub-base as a levelling/bedding course.

-----RCC RAMP SLAB-----

- I. 100 mm thick RCC slab (1:1.5:3), reinforced with 8 mm Ø bars @ 150 mm c/c in both directions as the main structural layer.
- II. Additional distribution reinforcement of 8 mm Ø bars @ 200 mm c/c provided along the length and along the width of the ramp.
- III. Overall ramp dimensions of approximately 2050 mm × 1200 mm, laid to a slope of 12.68° to connect the ground level to the entrance level.

-----FLOORING FINISH-----

- I. 20 mm thick cement mortar (1:4) bedding layer.
- II. 18 mm thick flamed (rough-finish) granite stone flooring over the bedding, provided over a width of approximately 900 mm, chosen specifically for its slip-resistant surface texture.
- III. A linear anti-skid strip pattern is provided on the finished surface, with strips set out at 200 mm × 200 mm spacing, to further improve grip for wheelchair users and pedestrians.

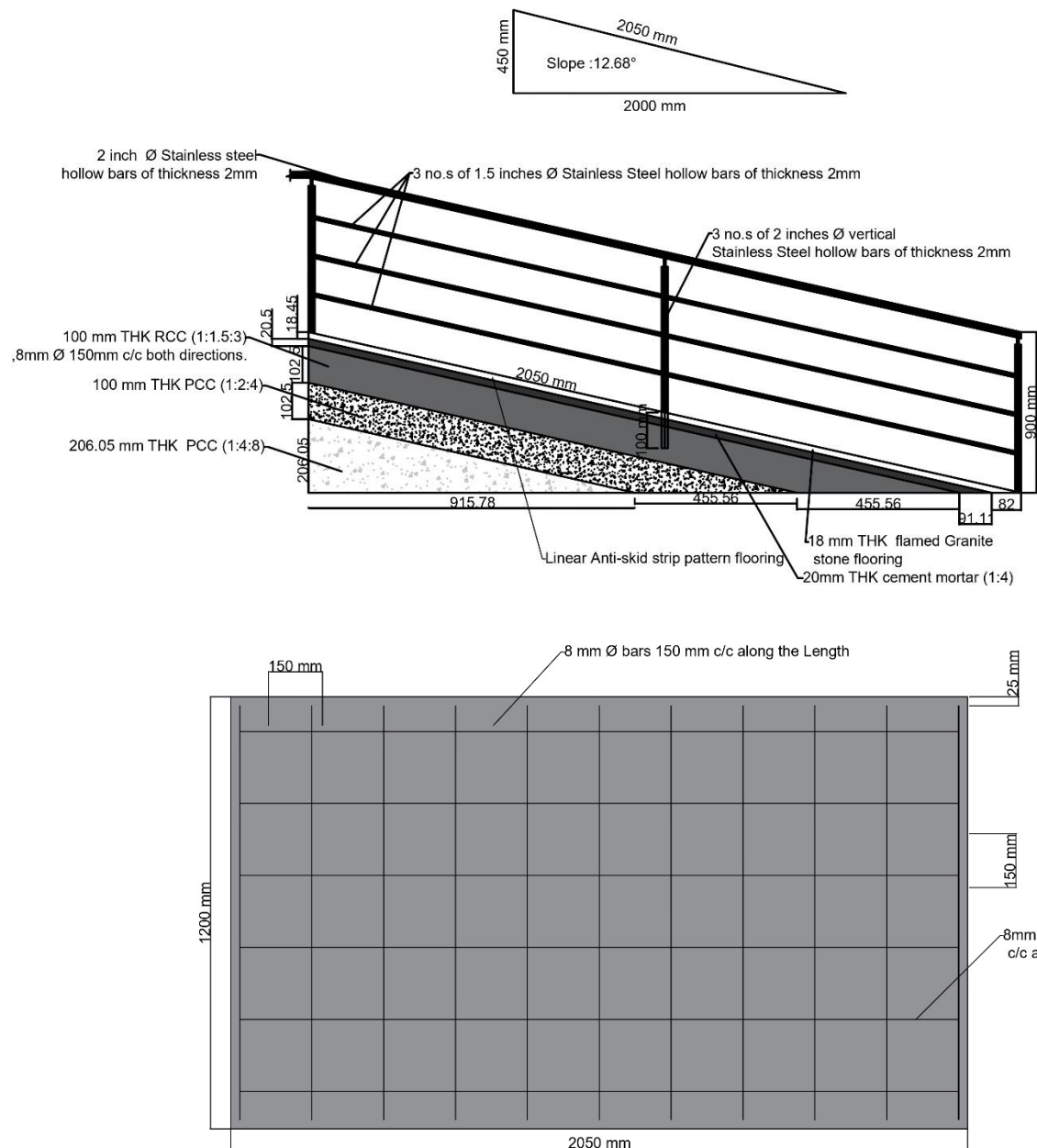
-----HANDRAILS-----

- I. 2-inch Ø SS hollow top handrail, 2 mm wall thickness.
- II. 3 nos. of 1.5-inch Ø SS hollow intermediate bars, 2 mm wall thickness.

- III. 3 nos. of 2-inch Ø vertical SS hollow support posts, 2 mm wall thickness, approximately 2000 mm high at 450 mm spacing.
- IV. The railing is provided along the ramp for the full run to give users a continuous handhold while ascending/descending.

3.3. ARCHITECTURAL & STRUCTURAL DRAWINGS

The following pages are the key architectural and structural drawings of the Ramp for the preparation of its **Quantity Estimation**, the **Bar Bending Schedule (BBS)** and **Bill of Quantities (BOQ)**.



3.4 METHODOLOGY

- **Step 1:** Reviewed the ramp drawing to note slope (12.68°), horizontal run (2000 mm), rise (450 mm), and slab length (2050 mm).
- **Step 2:** Calculated PCC quantities — 206.05 mm THK (1:4:8) base layer and 100 mm THK (1:2:4) leveling layer — using $\text{length} \times \text{width} \times \text{thickness}$.
- **Step 3:** Computed RCC slab quantity (100 mm THK, M-15/1:1.5:3) and reinforcement as per mesh detail — 8 mm $\varnothing$ bars @200 mm c/c both ways, 25 mm cover — for a 2050×1200 mm panel.
- **Step 4:** Measured flooring quantities — 18 mm THK flamed granite over 20 mm THK cement mortar (1:4) — including anti-skid strip allowance.
- **Step 5:** Worked out railing steel — 2" $\varnothing$ handrail, 3 nos. 1.5" $\varnothing$ mid-rails, and 3 nos. 2" $\varnothing$ vertical posts (900 mm ht) in stainless steel hollow section, 2 mm thick — converted to running length and weight.
- **Step 6:** Applied DSR-2023 rates for PCC/RCC/flooring items and market/LOA rates for SS railing, then compiled into BOQ.
- **Step 7:** Cross-verified quantities against the drawing and got them checked by the site engineer.

QUANTITY OF ESTIMATION

CALCULATION SHEET								
Name of work : Dismantling of the existing stairs and Construction of ramp thereof at the entrances of Control room building and Transit camp of 132/400 KV POWERGRID Silchar substation as per BoQ & instruction of Engineer-in-charge.								
			LENGTH	2	HEIGHT	0.45	WIDTH	1.2
S.NO	DSR'23 REF	ITEM DESCRIPTION	UNIT	NOS	MEASUREMENT			QTY
					LENGTH	BREADTH	HEIGHT	
1	15.23.1	DISMANTLING AND DEMOLISHING Dismantling already existing tile work of thickness between 10mm and 25mm on stairs	sqm					
		TREAD		6	1.2	0.75		5.4
		RISER		6	1.2	0.15		1.08
TOTAL:								6.480
2	5.9.1	FORM WORK Centering and shuttering including strutting,propping etc, and removal of form for foundations,footings,bases of columns,etc.for mass concrete.	sqm	4	2		0.412	1.648
3	4.1.8	PCC WORK Providing and laying cement concrete of 1:4:8 (1 Cement : 4 coarse sand (zone-III) derived from natural sources : 8 graded stone aggregate 40 mm nominal size derived from natural sources), excluding the cost of centering and shuttering-all up to plinth level.	cum	2	0.916	1.2	0.206	0.230
4	4.1.3	PCC WORK Providing and laying cement concrete of 1:2:4 (1 cement : 2 coarse sand (zone-III) derived from natural sources : 4 graded stone aggregate 20 mm nominal size derived from natural sources),excluding the cost of centering and shuttering - all upto plinth level.	cum	2	1.172	1.2	0.1	0.281

[illegible]

BILL OF QUANTITIES (BOQ)

BILL OF QUANTITIES (BOQ)									
Name of work : Dismantling of the existing stairs and Construction of ramp thereof at the entrances of Control room building and Transit camp of 132/400 KV POWERGRID Silchar substation as per BoQ & instruction of Engineer-in-charge.									
Item	Service Activity No.	DSR 23 Item Code No.	Description of Item	Unit	Quantity	DSR 2023 Rate (Rs.)	Rate Including C.I @ 44.51% (Rs.)	Rate excluding GST (Rs.)	Amount (Rs.)
					A	B	C=B x 1.445	D= C / 1.18	F= A x D
1	120001677	15.23.1	Dismantling tile work in floors and roofs laid in cement mortar including stacking materials within 50 metres lead, for thickness of tiles 10mm to 25 mm.	sqm	6.480	73.4	106.06	89.88	582.42
2	120000499	5.9.1	Centering and shuttering including strutting,propping etc, and removal of form for foundations,footings,bases of columns,etc.for mass concrete.	sqm	1.648	392.15	566.66	480.22	791.40
3	120000442	4.1.8	Providing and laying cement concrete of 1:4:8 (1 Cement : 4 coarse sand (zone-III) derived from natural sources : 8 graded stone aggregate 40 mm nominal size derived from natural sources), excluding the cost of centering and shuttering-all up to plinth level.	cum	0.230	6812.0	9843.34	8341.81	1918.62
4	120000437	4.1.3	Providing and laying cement concrete of 1:2:4 (1 cement : 2 coarse sand (zone-III) derived from natural sources : 4 graded stone aggregate 20 mm nominal size derived from natural sources),excluding the cost of centering and shuttering - all upto plinth level.	cum	0.281	7878.5	11384.43	9647.82	2711.04

[illegible]

4. BAMBOO CRIB WALL

(A Vegetative Retaining Structure for Slope & Embankment Protection)



4.1 INTRODUCTION

As part of my internship at POWERGRID, I studied the construction methodology and engineering principles of the **Bamboo Crib Wall**, a bio-engineering solution adopted for slope and embankment protection along transmission line corridors. This report summarises the concept, design parameters, and step-by-step construction procedure of bamboo crib walls, as documented in the project's Standard Operating Procedure (SOP).

4.2 WHAT IS A CRIB WALL?

A crib retaining wall is a gravity-retaining structure built up of individual box-like cells called cribs, formed by alternating **header** and **stretcher** elements. Each cell is filled with on-site soil, which acts as the primary retaining mass, while the crib framework holds the fill in place and resists overturning caused by slope pressure. Crib elements can be made of concrete, timber, or bamboo. When bamboo is used together with live plant cuttings placed between the cells, the structure is termed a **live/vegetative bamboo crib wall** - a low-cost, eco-friendly alternative to conventional retaining walls, well suited for hilly and erosion-prone terrain such as that found in Assam and other North-Eastern project sites.

Depending on the material and arrangement of crib elements, four basic crib configurations (Type 1 to Type 4) are possible, and walls up to 25 m in height can theoretically be built in steps as per site requirement. For this project, low-height Type 1 or Type 2 bamboo cribs (1.0 m - 1.5 m high) are used, arranged in a series of 3-4 stepped walls up the slope. **Bhaluka**

bamboo, 8-10 m long and about 8-10 cm in average diameter, is preferred; where required, bamboos are bundled in threes for added strength.

4.3 CONSTRUCTION PROCEDURE (AS PER SOP)

- I. **MATERIAL PROCUREMENT:** Suitable bamboo is selected, cut, and transported to site along with soil/earth sourced locally (recorded via challan as a minor mineral). A joint visual inspection confirms use of Bhaluka bamboo before further lopping.
- II. **LAYOUT & MARKING:** The site is jointly measured and demarcated for the wall alignment.
- III. **BINDING MATERIAL:** Scrap conductor from the PGCIL store is used for binding the bamboo members.
- IV. **SEQUENCE:** To prevent overturning, construction always starts from the bottom-most portion of the slope/embankment and proceeds upward.
- V. **EMBEDMENT:** The crib wall is founded at least 400 mm below the existing ground level.
- VI. **PEGGING:** Bamboo pegs are driven vertically 1.5 m into the ground (0.6 m projecting above ground) at every 1.5 m along the stretcher length - 25 pegs for a 35 m wall and 13 for an 18 m wall, staggered between adjacent cribs to avoid alignment along the same line.
- VII. **BASE LAYER:** The bottom-most crib layer is laid with header bamboos, topped by stretcher bamboos, since the shorter header span resists deflection of the stretchers above.
- VIII. **WALL GEOMETRY:** An overall width of 1.5 m (top to bottom) is maintained with a slope of approximately 30 deg through the height; bamboo bundles are cut with a 0.5 m lap allowance and tied with scrap conductor to the required stretcher lengths (35 m / 18 m walls).
- IX. **BACKFILLING:** Soil is filled inside the crib cells in 0.2 m layers, watered, and compacted manually with a rammer along the full stretcher length before the next layer begins.
- X. **SUCCESSIVE LAYERS:** Each subsequent layer is placed with a batter (deflection) of 15 deg from the layer below, continuing till the design height of 1.5 m is reached.
- XI. **BIO-STABILISATION:** Live cuttings/stems of fast-growing species are laid horizontally between layers at 15-20 cm spacing; vetiver, lemon grass, and dwarf bamboo are planted on the slope face to control erosion, timed to take advantage of the monsoon season.
- XII. **DOCUMENTATION:** Geo-tagged photographs are maintained for all major activities and shared with POWERGRID as required.

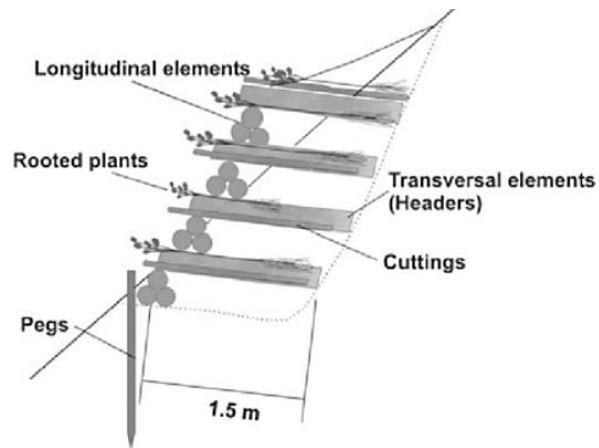


Fig. 1: Schematic of a bamboo crib wall showing longitudinal elements, transversal headers, rooted plant cuttings, and anchoring pegs

4.4 KEY DESIGN PARAMETERS

PARAMETER	SPECIFICATION
BAMBOO SPECIES	Bhaluka bamboo
BAMBOO LENGTH / DIAMETER	8-10 m long, ~10 cm average diameter
WALL HEIGHT (PER TIER)	1.0 m - 1.5 m (3-4 tiers up the slope)
WALL WIDTH (TOP TO BOTTOM)	1.5 m (typ.); 1.0-1.2 m per SOP note
EMBEDMENT BELOW GROUND	Min. 400 mm
HEADER	Will be laid horizontally along the width of the crib wall & inserted min 1 m into the ground

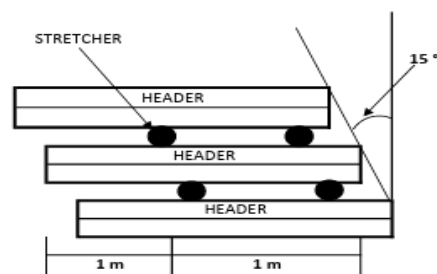
STRETCHER	Will be laid horizontally along the length of the crib wall & min 0.5 m lapping shall be considered
PEG LENGTH	1.5 m into ground + 0.6 m above ground
PEG SPACING ALONG STRETCHER	1.5 m (25 nos./35 m wall, 13 nos./18 m wall)
BACKFILL COMPACTION LAYER	Maximum 0.2 m at a time, watered and rammed manually
BATTER / SLOPE THROUGH HEIGHT	~30 deg overall; 15 deg per successive layer
LIVE CUTTING SPACING	15-20 cm, laid horizontally between layers
LAP ALLOWANCE FOR BAMBOO JOINTS	0.5 m

3.5 WORKING PRINCIPLE & ADVANTAGES

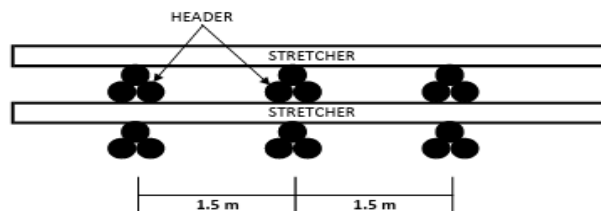
The bamboo initially provides immediate mechanical support to the backfill while the embedded live cuttings begin rooting into the soil. As the exposed portions of the cuttings sprout leaves, the structure develops a natural vegetative cover, and the growing root mass gradually binds the fill and parent soil into a single coherent mass. Over time, the role of the bamboo (which eventually decomposes) is taken over by this root network, permanently stabilising the slope. The method is therefore low-cost, uses locally available and biodegradable material, blends aesthetically with the surrounding landscape, and needs no imported gardening effort once the cuttings establish themselves.



Fig. 2: Construction stages of a bamboo crib wall - excavation and layout, layered bamboo cribs with backfill, and the completed vegetated wall



Bamboo Crib Wall (Side View)



Bamboo Crib Wall (Front View)

Bamboo Crib Wall for Loc 150 of 400kV D/C Silchar – Imphal Transmission Line

5. ABOUT THE PROJECT

-----PROJECT NAME-----

Structural Design and Drawing of Roof Shed for Crane with combination of RCC & Steel Structures at the 132/400 kV POWERGRID Silchar Substation Premises

The Crane Shed is a proposed steel-and-brick structure to be constructed within the substation premises, intended to serve as a dedicated shelter for the crane used during equipment handling, erection, and maintenance activities at the substation. Cranes and associated lifting equipment are essential assets for the installation and upkeep of heavy electrical apparatus such as transformers, circuit breakers, and other switchyard equipment; housing the crane in a purpose-built shed protects it from direct exposure to rain, sunlight, and dust, thereby reducing corrosion, preserving the working condition of its mechanical and electrical components, and extending its service life



The shed is designed as an open-sided portal-frame structure with a pitched pipe-truss roof, short brick knee walls, and an entry ramp, allowing the crane to be driven in and parked with ease. The layout follows a regular structural grid — four bays of 3.25 m each along the length (gridlines 1 to 5) and a single bay of 5.5 m along the width (gridlines A to B) — giving an overall plan footprint of approximately 13.0 m x 5.5 m. This grid-based, modular approach keeps the structure simple to fabricate, cost-effective, and quick to erect, while providing adequate clear height and span for crane movement and parking.

The project scope covers the complete structural package for the shed: RCC foundations and footings, structural steel columns (posts) with base connections, a lattice pipe-truss roof structure, brick infill walls, and a vehicular access ramp with its own RCC/PCC detailing. The following section describes the structural system and the specifications adopted for each component, as derived from the design and detail drawings.

5.1. SITE AND CONNECTIVITY

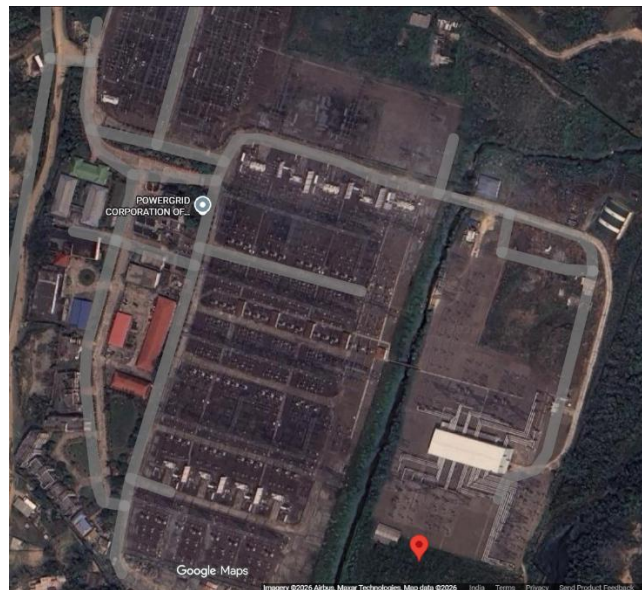


Figure 4 : Google Map image of PGCIL,Srikona

The geographic and logistical placement of the site plays a critical role in its construction supply chain (procurement of aggregate, cement, and steel).

- **Latitude & Longitude:** 24°49'34.0"N 92°43'19.8"E
- **Location:** Power Grid 400/132 kV substation, Srikona, Assam
- **Connectivity:** The site is connected directly to two lane National Highway (NH 37) that serves as a strategic transportation corridor linking Imphal (Manipur) to Silchar and Sribhumii in Assam, via Lalmati TV Tower Road at Indragarhgrant, ensuring all weather efficient connectivity. Its proximity to major transit corridors ensures the unhindered movement of heavy machinery, Transit Mixers, and Ready Mix Concrete (RMC) vehicles.

5.2 STRUCTURAL DESCRIPTION AND SPECIFICATIONS

5.1.1 OVERALL LAYOUT AND GRID

The shed is planned on a rectangular column grid as follows:

Parameter	Specification
GRID LINES (LENGTH)	1 - 2 - 3 - 4 - 5 (4 bays)
BAY SPACING (LENGTH)	3.25 m c/c x 4 bays = 13.0 m
GRID LINES (WIDTH)	A - B (single bay)
BAY SPACING (WIDTH)	5.5 m c/c
OVERALL PLAN FOOTPRINT	≈ 13.0 m x 5.5 m
OVERALL HEIGHT (FGL TO ROOFING/RIDGE)	5.0 m
FOUNDATION LEVEL (BELOW FGL)	-2.1 m
BRICK WALL (KNEE WALL) LEVEL	1.2 m above plinth

5.1.2 FOUNDATION AND FOOTING

Each structural steel post is supported on an isolated RCC footing. The footing is founded 2.1 m below finished ground level (FGL), well below the brick-wall plinth level, to ensure

adequate bearing and stability against the lateral and uplift loads transferred from the portal frame and roof truss.

Parameter	Specification
FOOTING DEPTH	1.65 m (1650 mm)
BLINDING LAYER	75 mm PCC (1:4:8) with 75 mm projection all around
MAIN REINFORCEMENT	12 mm dia bars @ 150 mm c/c both ways
COLUMN STARTER BARS	4 - 16 mm dia & 4 - 20 mm dia
STIRRUPS	8 mm dia @ 175 mm c/c
ADDITIONAL BARS	6 - 12 mm dia with 8 mm dia ties @ 150 mm c/c
REFERENCE LEVELS	NGL (Natural Ground Level) and PL (Plinth Level) as datum

5.1.3 COLUMNS (STRUCTURAL STEEL POSTS)

Vertical support is provided by ISMB 150 steel posts (Indian Standard Medium Weight Beam, 150 mm depth), bolted to the foundation through a fabricated base-plate assembly and connected at the top to the roof truss through welded support plates.

Parameter	Specification
POST SECTION	ISMB 150

Parameter	Specification
BASE PLATE	250 x 250 x 10 mm thick MS plate, bolted with 4 nos. 20 mm dia x 500 mm long anchor bolts
TRUSS SUPPORT PLATE (TOP)	150 x 150 x 8 mm MS plate welded to ISMB 150
BRACING PLATE	125 x 125 x 8 mm MS plate welded to ISMB 150
SECONDARY PLATE	150 x 75 x 8 mm MS plate welded to ISMB 150
CHANNEL	ISMC 100 welded to plate
ANGLE BRACING	ISA 50x50x6 and ISA 30x30x5, welded to plate
RCC COLUMN (RAMP/PLINTH, IF APPLICABLE)	400 mm x 400 mm with 2-20 mm / 4-16 mm dia main vertical bars and 8 mm dia stirrups @ 175 mm c/c
RCC BEAM	300 mm x 400 mm with 6-12 mm dia main bars and 8 mm dia stirrups @ 150 mm c/c

5.1.4 ROOF TRUSS (LATTICE PIPE STRUCTURE)

The roof is formed by a lattice-type pipe truss spanning the 5.5 m width, pitched to a central ridge to shed rainwater to both sides. The truss is fabricated from mild-steel (MS) pipes of varying diameters graded by load path — heavier pipes at the principal chords and lighter pipes for the lattice/lacing members — and is supported directly on the ISMB 150 posts through the welded plate connections described above.

Parameter	Specification
PRINCIPAL (HEAVY) CHORD MEMBER	80 mm NB MS pipe (H)
SECONDARY CHORD / RAFTER MEMBER	65 mm NB MS pipe (M)
LACING / LATTICE MEMBERS	50 mm NB MS pipe (M)
PURLIN SPACING (BOTH SIDES)	4 nos. 50 mm NB MS pipes @ 1.1 m c/c
TRUSS BOTTOM MEMBERS	10 nos. 50 mm NB MS pipe (M) @ 1.44 m c/c
SPAN (EAVE TO EAVE)	5.5 m
EAVE OVERHANG	0.5 m
RAFTER LENGTH (EAVE TO RIDGE)	3.286 m
ROOF PITCH HEIGHT REFERENCES	1.8 m and 1.375 m at intermediate purlin lines

5.1.5 BRICK WALLS

Short knee walls in 230 mm (0.23 m) thick brick masonry are provided between the steel posts up to a height of 1.2 m above plinth level, forming a solid dwarf wall that stiffens the base of the structure, protects the interior from splash water and debris, and provides a visually finished plinth all around the open-sided shed.

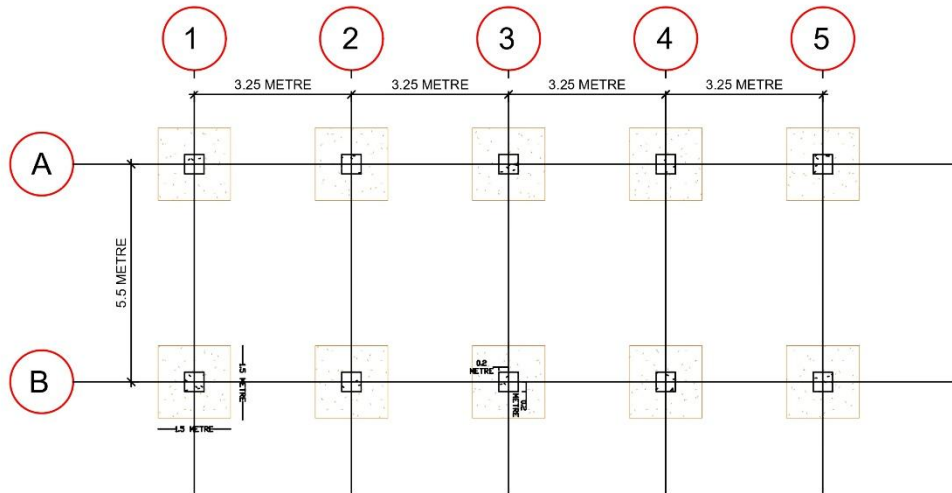
5.1.6 ENTRY RAMP

A ramp is provided at the shed entrance to allow the crane to move smoothly from the existing ground level (EGL) up to the finished ground level (FGL) / plinth of the shed.

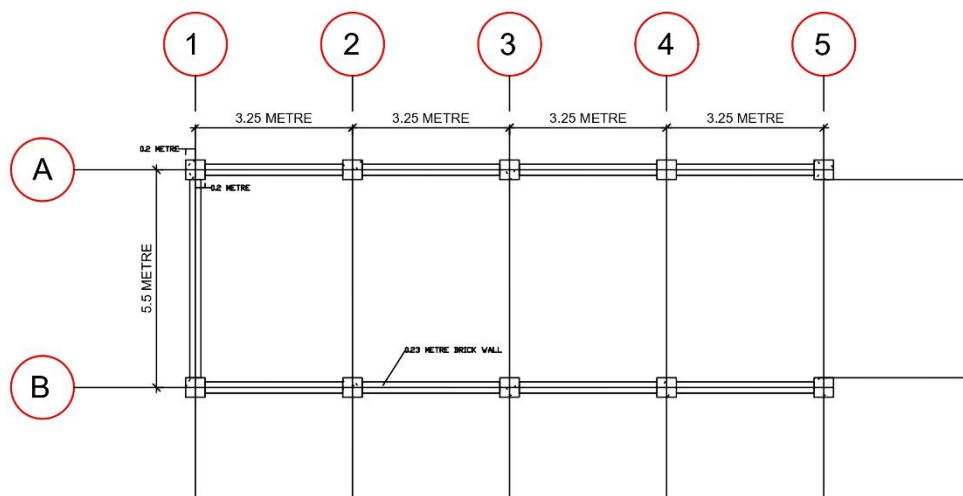
Parameter	Specification
RAMP LENGTH	3000 mm (3.0 m)
BASE LAYER	100 mm sand filling
MASONRY LAYER	150 mm full brick work
RCC SLAB	100 mm thick RCC (1:1.5:3) with 8 mm dia bars @ 300 mm c/c both ways
BLINDING	75 mm PCC (1:4:8)
REFERENCE LEVELS	EGL (Existing Ground Level) and FGL (Finished Ground Level)

5.1.7 ELEVATIONS

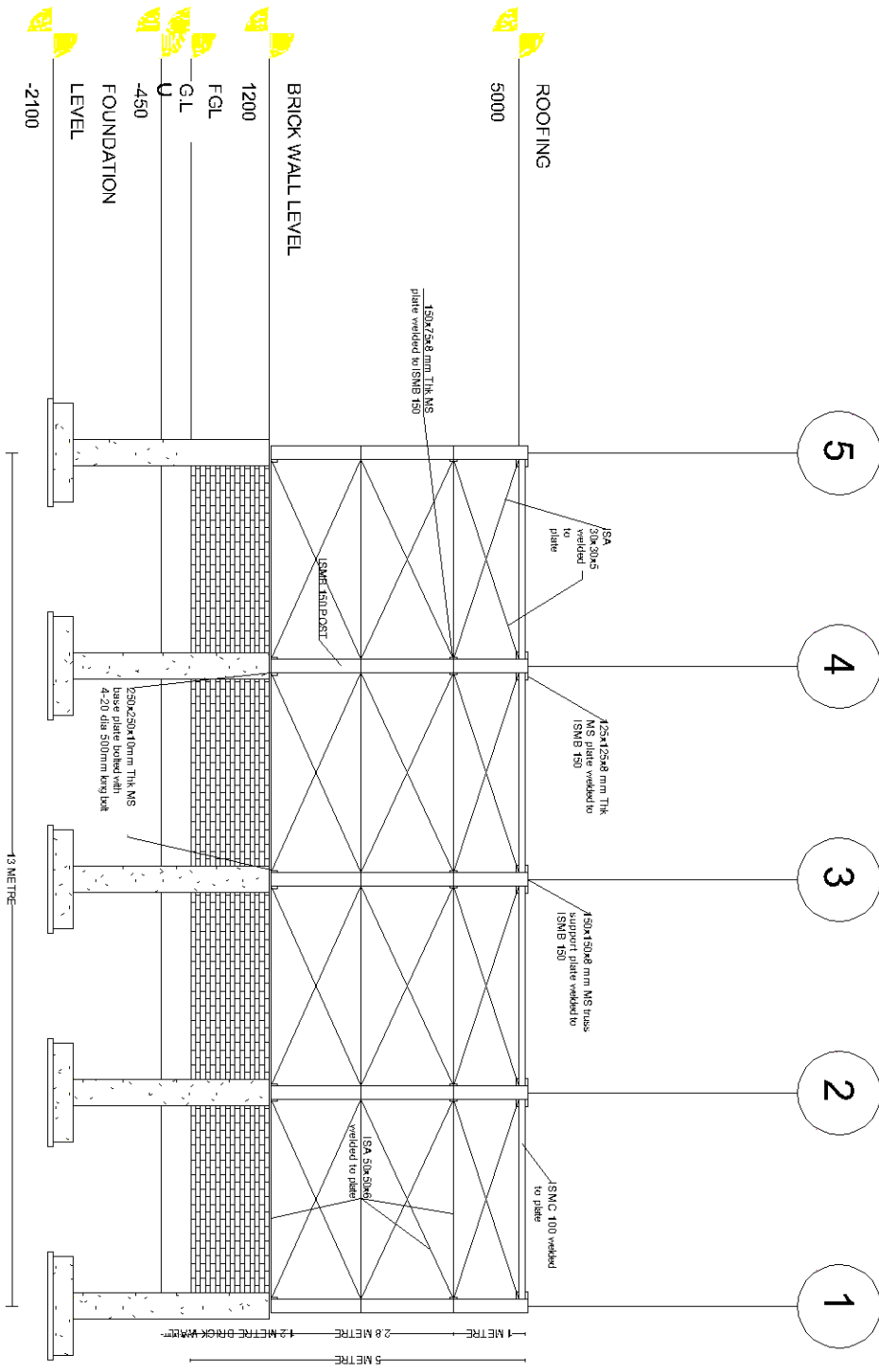
Front and rear elevations show a symmetrical portal profile: a 1.2 m high brick knee wall rising from a 0.45 m plinth, steel posts rising to a 5.0 m eave/roofing level, and the pitched pipe-truss roof above, with a 1.8 m clear zone below the eave line and a 0.5 m eave overhang on both the front and rear faces (over the 5.5 m span).

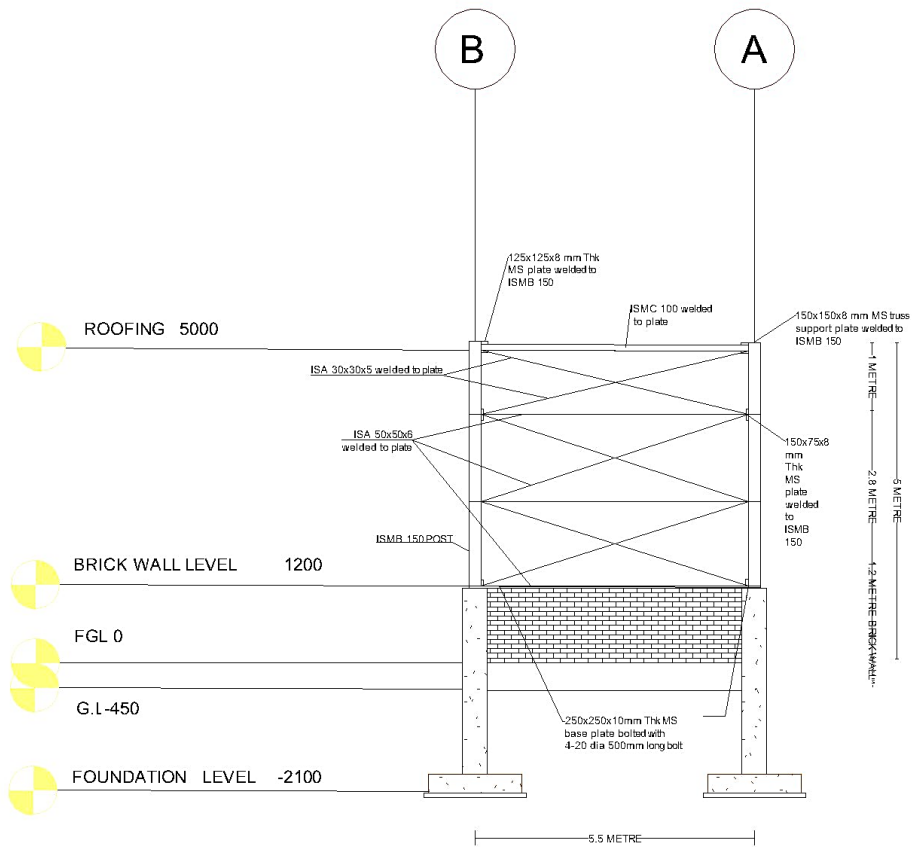
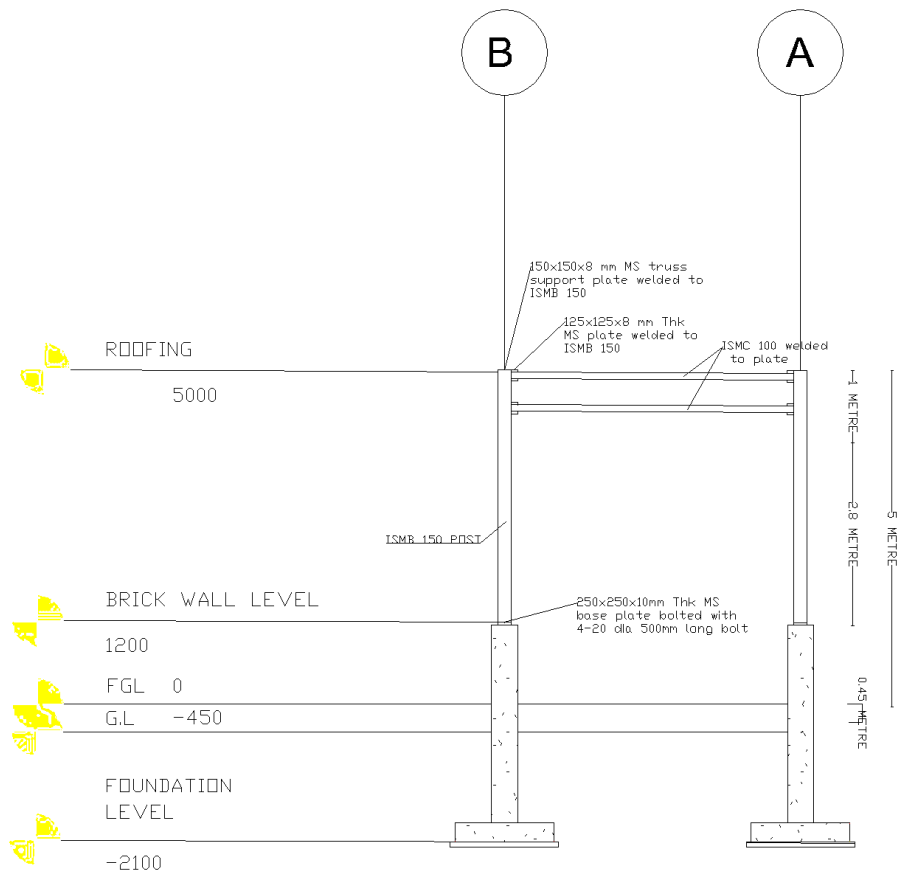


RCC DETAILS OF FOOTING

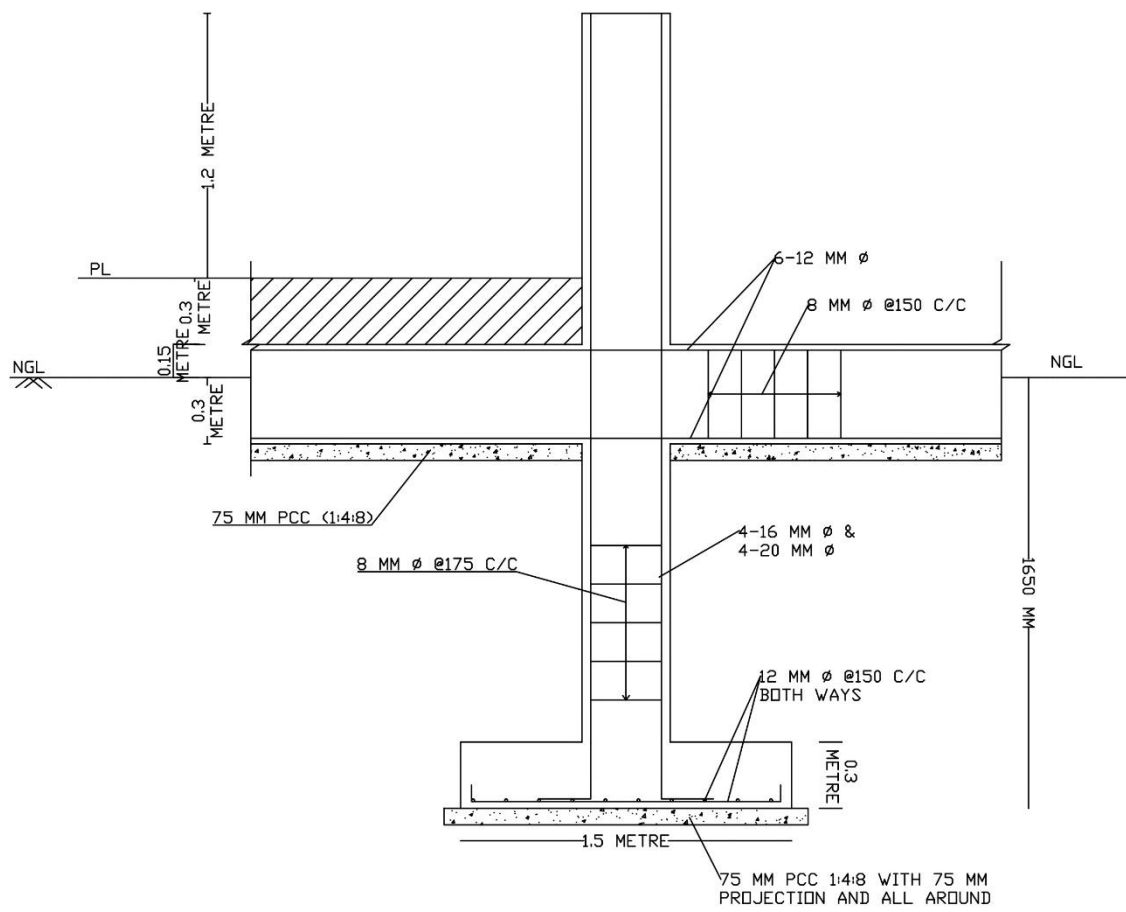


COLUMN-WALL PLAN

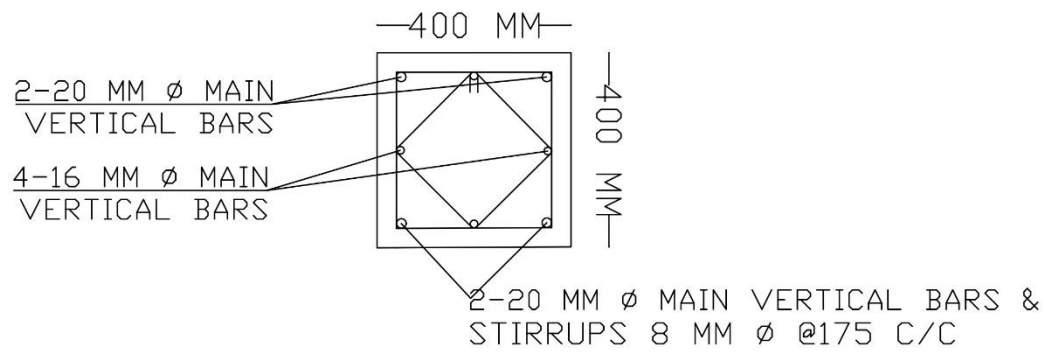




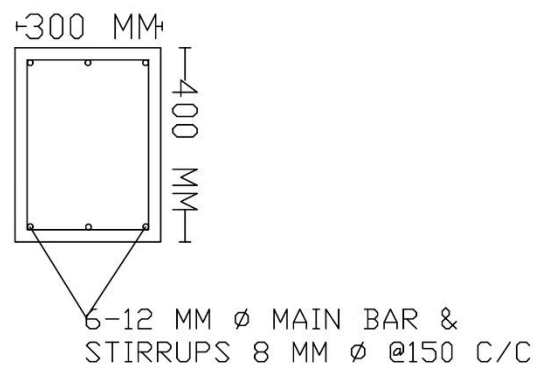
LATTICE STRUCTURE DETAILS (VIEW FROM INNER SIDE)



SECTIONAL DETAILS OF FOOTING



COLUMN CROSS SECTION



BEAM CROSS SECTION

4-50 MM NB MS PIPES @1.1 METRE C/C AT BOTH SIDES

65 MM NB MS PIPE (M)

3.286 METRE

0.5 METRE

1.375 METRE

5.5 METRE

1.375 METRE

RIDGE

10-50 MM NB MS PIPE (M) @1.44METRE C/C

50 MM NB MS PIPE (M)

18 METRE

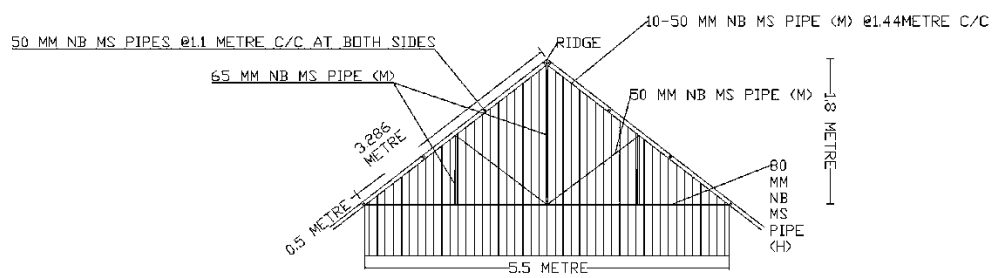
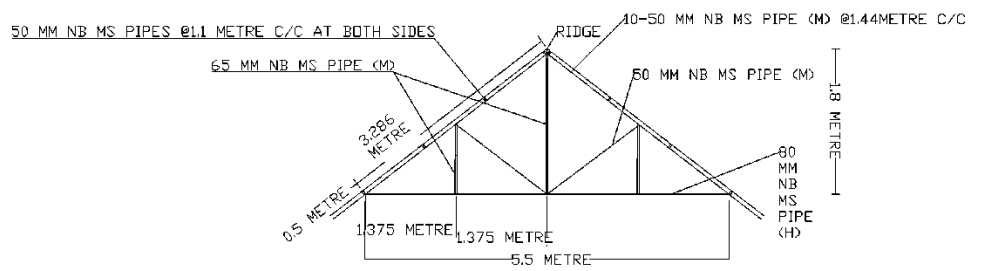
5 METRE

1.2 METRE

0.45 METRE

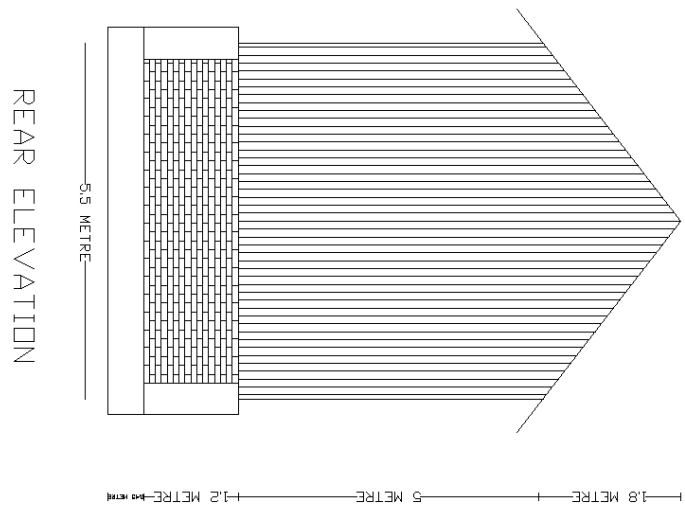
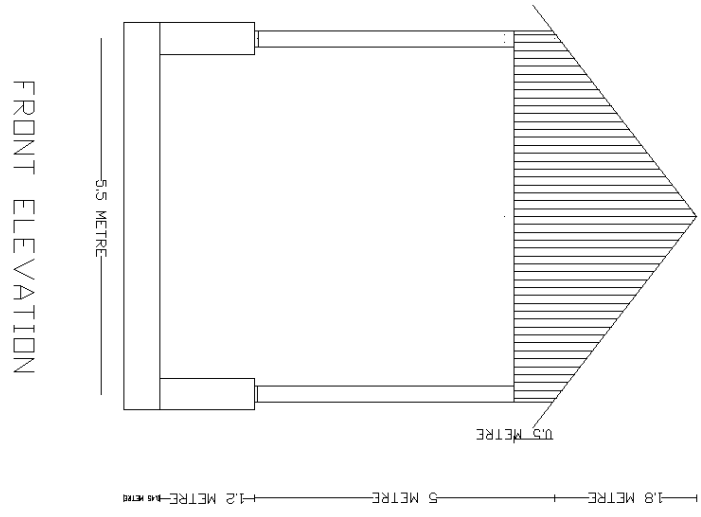
1.650 METRE

80 MM NB MS PIPE (H)



TYPICAL SECTIONAL DETAILS

SSSS



5.1. WORKFLOW OF PROJECT EXECUTION

In addition to the technical execution of individual works, every civil works contract at POWERGRID follows a defined administrative and site-management procedure, from the pre-award stage through to final contract closing. This works procedure, as observed and followed during the projects undertaken in this internship, is summarised below.

A. PRE-AWARD ACTIVITIES

- **COST ESTIMATE APPROVAL** : Technical(Drawing, BoQ, suitability of site, its connectivity) and administrative approval of the cost estimate is taken. This whole thing goes under several cross check to avoid unnecessary costing by officers at designated levels.
- **TENDERING** : Tendering of the work come in process, depending on the value and importance of the work number of competing vendors are decided. For small work usually trusted, well known three competitors are given chance during work awarding. Whereas for big projects, the scale of competing vendors are extended and are often given on the basis of trust, work experience and Bid Estimates.

B. POST-AWARD ACTIVITIES

- **DOCUMENT REVIEW** : Review of documents as per the Work Order / Letter of Award (LOA). This contains necessary information and conditions which mainly stands with and protects the client at every phase during the construction. This mainly includes:

Work Completion Period

Quantity Variation

Liquidated Damages for delay

Defect Liability

Insurance

Third Party Liability

Workmen Compensation Policy

Terms of Payment

Mode of Payment

Payment Tracking**Settlement of disputes and Abbitration****Social Accountability**

- **MATERIAL TESTING** : Material testing as per the Standard Field Quality Plan (SFQP) and quality control
- **SITE SUPERVISION** : Site supervision as per approved drawings, along with maintenance of all applicable registers
- **BILLING** : Raising and processing of running/final bills
- **FINAL AMENDMENT** : Final amendment to the work order after completion of work, if required

C. CONTRACT CLOSING

- **RECOVERIES** : Finalisation of all recoveries due under the contract
- **RETENTION RELEASE** : Release of the extra retention amount withheld during execution

This sequence – Pre-Award Activities, Post-Award Activities, and Contract Closing – formed the administrative backbone within which the technical and safety practices described in the following section were carried out.

6. SAFETY PRECAUTION FOR ENGINEERS DURING CONSTRUCTION

The civil works undertaken during this internship - the underground RCC bunker, the accessibility ramp, the bamboo crib slope protection, and the crane shed - involved deep excavation, reinforced concrete work, work at height, and construction in the immediate vicinity of a live 132/400 kV switchyard. Such an environment demands rigorous adherence to safety practice by every engineer engaged in supervision and execution. The following safety requirements were observed to be essential on site and are relevant to civil engineers working on similar building construction projects at power transmission installations.

6.1 PERSONAL PROTECTIVE EQUIPMENT (PPE)

Every engineer and worker on site must use ISI-marked safety helmets, high-visibility jackets, safety shoes with steel toe caps, and gloves appropriate to the task. Safety harnesses and lanyards are mandatory during staircase, mumty, and roof-level RCC work above 1.8 m, and dust masks and ear protection are required near concrete mixing, cutting, and bending operations.

6.2 EXCAVATION AND FOUNDATION SAFETY

The bunker raft was founded at (-)3.15 m, requiring deep open excavation. Engineers must ensure side slopes or shoring/shuttering are provided to prevent soil collapse, that excavated soil is stacked at a safe distance from the edge, and that barricading, warning signage, and adequate lighting are provided around open pits. Dewatering arrangements must be checked before workers enter the excavation, and a competent person should inspect the excavation daily before work resumes.

6.3 WORKING AT HEIGHT

Construction of the mumty up to (+)3.15 m, the staircase flights, and the crane shed roof truss at 5.0 m eave level involves sustained work at height. Engineers must ensure scaffolding is erected on firm base plates, is fully boarded with toe boards and guard rails, and is tagged and inspected before use. Full-body harnesses anchored to independent lifelines are mandatory for roof truss erection and edge work, and ladders must be secured and used at the correct angle.

6.4 ELECTRICAL SAFETY NEAR LIVE SUBSTATION EQUIPMENT

Since all four projects are located within the operational 132/400 kV Srikona substation, civil engineers must maintain the statutory minimum approach distance from live equipment and busbars at all times, and no crane, scaffold pipe, or reinforcement bar may be handled or stored within the safe clearance zone of energised conductors. A Permit to Work must be obtained from the substation's operation and maintenance staff before starting work in any switchyard bay, and work must stop immediately if there is any doubt about proximity to live parts.

6.5 REINFORCEMENT, CONCRETING AND MATERIAL HANDLING

Bar bending and cutting must be carried out only by trained operators using guarded machines, with sharp bar ends capped to prevent injury. During concrete pours, engineers must ensure vibrator cables are earthed, batching and mixing areas are kept clear of loose material, and Transit Mixer and RMC vehicle movement on site is directed by a trained signaller to avoid collision with workers and existing structures. Manual lifting of steel and formwork should follow safe lifting technique, with mechanical aids used for loads beyond manual handling limits.

6.6 CRANE AND LIFTING EQUIPMENT SAFETY

As the crane shed is designed to facilitate the safe parking and maintenance of cranes, engineers must ensure cranes entering via the entry ramp operate on firm, level ground with load charts followed strictly, that the outrigger zone is barricaded, and that no lifting operation is carried out beneath or near overhead transmission lines without prior clearance and an observer for boom proximity to live conductors.

6.7 APPLICABLE CODES AND REGULATORY FRAMEWORK

Site safety practice was guided by IS 3764 (Safety Code for Excavation), IS 7969 (Safety Code for Handling and Storage of Building Materials), the National Building Code of India Part 7 (Construction Management, Practices and Safety), the Building and Other Construction Workers (BOCW) Act, 1996, and the Central Electricity Authority's safety regulations applicable to substation premises. CPWD Specifications 2019 also prescribe site safety and welfare provisions that were followed alongside these codes.

Together, these requirements reflect that safety on a power transmission construction site is inseparable from good engineering practice: the site engineer is responsible not only for the structural integrity of the works but also for the wellbeing of every worker engaged in their execution.

7. CONCLUSION

The 45-day summer internship at Power Grid Corporation of India Limited, Srikona, provided valuable hands-on exposure to civil engineering practice on a live, operational 132/400 kV transmission substation. Working under the guidance of the site engineers, the internship covered the design interpretation, quantity estimation, Bar Bending Schedule preparation, and Bill of Materials documentation for four distinct upcoming infrastructure works: the underground RCC protective bunker for disaster preparedness, the accessibility ramp at the Transit Camp and Control Room, the bamboo crib wall for slope and embankment protection, and the structural design of the crane shed.

Each project offered a different dimension of civil engineering application. The bunker required detailed understanding of RCC detailing for a partially below-ground structure, including raft foundation design, wall and column reinforcement, and staircase and masonry construction, all executed in accordance with IS 456, IS 875, IS 1893, IS 2502, and IS 13920. The wheelchair access ramp highlighted the importance of inclusive design in public infrastructure, while the bamboo crib wall demonstrated the use of sustainable, locally sourced materials for slope stabilisation and erosion control around vulnerable transmission tower locations. The crane shed, with its steel pipe truss roof and RCC entry ramp, gave exposure to composite structural systems combining steelwork and reinforced concrete.

Beyond the technical design and estimation work, the internship offered practical insight into tender-stage documentation, CPWD Delhi Schedule of Rates-based cost estimation, and the specifications that govern quality and execution on government infrastructure projects. Working within an active substation environment also underscored the critical importance of construction safety, particularly the coordination required between civil works and electrical operations, and the discipline needed to execute excavation, working-at-height, and material handling activities safely near live equipment.

Overall, this internship successfully bridged classroom learning in reinforced concrete design, structural analysis, and quantity surveying with the realities of field practice on critical national infrastructure. It strengthened technical understanding of RCC and steel detailing, quantity estimation, and construction safety, and reinforced the professional responsibility that civil engineers carry in delivering infrastructure that is safe, durable, and fit for its intended purpose.

The experience gained at POWERGRID, Srikona, will serve as a strong foundation for future academic and professional pursuits in civil engineering.